# International Power

How asymmetric trade dependence becomes geopolitical leverage

Paper by Ernest Liu (Princeton) & David Y. Yang (Harvard)

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Presented by Victor Xucheng Chen

# The paper in one slide

**Idea.** If country $n$ buys many hard-to-substitute goods from $i$—more than $i$ buys from $n$—then $i$ can *hold* $n$ *up* by threatening to stop exporting. That asymmetry is **international power**.

**Q1 Consequences.** Does power trigger bilateral engagement and negotiation?
⇒ **Yes:** more power asymmetry today ⇒ more diplomatic engagement next year; UN votes converge.

**Q2 Causes.** Do countries build power when relations sour?
⇒ **Yes:** when a partner's close election worsens bilateral alignment, the other country raises its power over that partner—mostly by *cutting imports* from it.

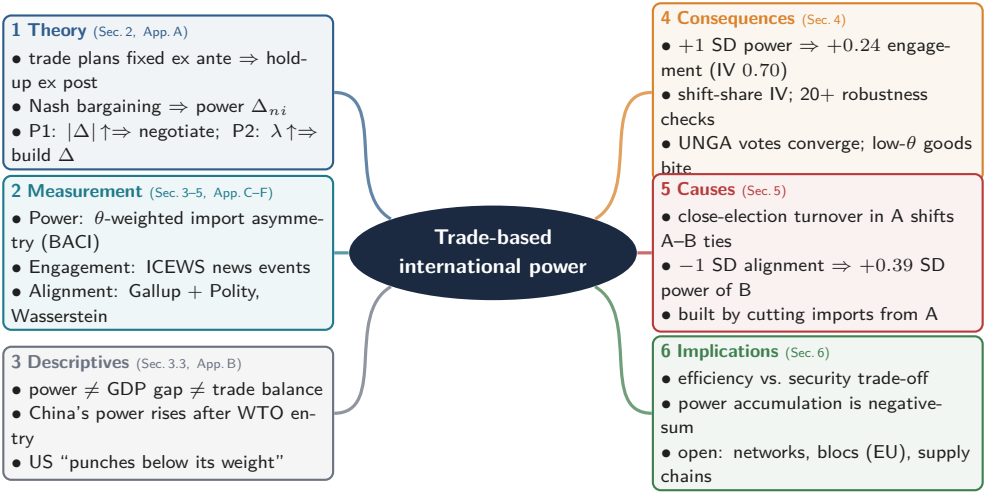
**+0.236**
engagement per +1 SD
power (IV: 0.697)

**+0.39**
SD of B's power over A
per −1 SD alignment

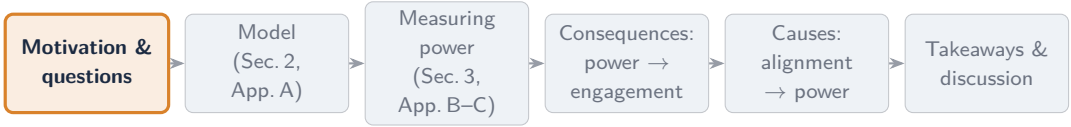
**+38.4%**
engagement at the mean
(+18.8% at median)

**0.96**
corr. of the measure with a full GE version

# Paper map: how the pieces fit together



# Part 1 — Motivation and questions




- Historical cases where trade was used as a weapon
- What exactly the paper measures and asks
- Where it sits in the literature

# Trade can be weaponised

**1973 oil embargo.** During the Yom Kippur War, Saudi Arabia and other Arab producers embargoed the US over its support for Israel.

- US imported $\approx 35\%$ of its oil from OPEC members
- Arab producers $\approx 52\%$ of world oil production

**2010 critical-minerals halt.** In a fishing-rights dispute, China halted critical-mineral exports to Japan (inputs for hybrid cars, wind turbines, guided missiles).

- Japan sourced $\approx 87\%$ of its critical-mineral imports from China
- China controlled $\approx 97\%$ of world production and exports

## Hirschman (1945)

Trade creates relationships of dependence, influence and even domination. Two sources of power:

1. **asymmetry** in dependence
2. **cost of disruption** (how hard it is to substitute)

Every sovereign can interrupt—or threaten to interrupt—its own exports and imports.

**Takeaway.** The paper is about the *threat* of disruption, not realised sanctions: only 668 (5.3%) of country pairs ever sanctioned each other in the sample period.

# What the paper does

### 1. Theory

Ex-ante trade plans, ex-post disputes, hold-up.

$\Rightarrow$ a **measure** of power: asymmetric import dependence, weighted by $1/\theta_k$.

$\Rightarrow$ **two predictions**: shocks to power change engagement; shocks to alignment change power.

### 2. Three new measures

**Power**: all directed pairs, 2001–2021 (BACI trade, trade elasticities).

**Engagement**: ICEWS news-coded bilateral events (6,733,036 events, 23,516 pairs).

**Alignment**: Gallup leadership disapproval + Polity, located by "GPS-style" multilateration.

### 3. Causal evidence

**Consequences**: power $\rightarrow$ engagement. Within-pair TWFE + shift-share IV.

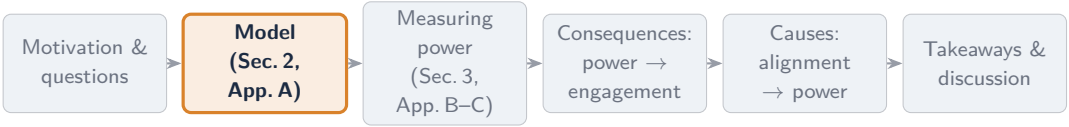
**Causes**: alignment $\rightarrow$ power build-up. Close-election turnover in a partner as an instrument.

**Mechanisms**: sectors, exporter vs. importer power, trade policies.

# Where the paper sits

| Literature | Representative work | What this paper adds |
|---|---|---|
| **Geoeconomics (theory)** | Hirschman (1945); McLaren (1997); Farrell & Newman (2019); Clayton, Maggiori & Schreger (2024, 2025); Becko & O'Connor (2025); Becko, Grossman & Helpman (2025); Thoenig (2023); Mayer, Mejean & Thoenig (2025) | first empirical evidence on *consequences* and *causes* of trade-based power |
| **Political economy of IR** | military (Gennaioli & Voth 2015); aid (Kuziemko & Werker 2006; Nunn & Qian 2014); covert ops (Berger et al. 2013); trade & war (Martin, Mayer & Thoenig 2008) | power from trade, beyond volume; peaceful and non-peaceful engagement |
| **Domestic ↔ foreign politics** | Grossman & Helpman (1994); Fearon (1994); Antràs & Padró i Miquel (2011) | domestic turnover → alignment → *other* countries' trade policy |
| **Measurement** | friends & enemies (Kleinman, Liu & Redding 2022); UNGA ideal points (Bailey et al. 2017); geopolitical risk (Caldara & Iacoviello 2022) | bilateral power, engagement, high-frequency alignment |

# Part 2 — An illustrative model of trade with disputes




- Why ex-post hold-up turns *import dependence* into power
- The power formula $\Delta_{ni}$ and the two predictions that the empirics test
- Why the shortcuts in the model do not drive the measure (Appendix A)

# Environment

- $N$ countries, $K$ sectors; a consumer, competitive firms and a government that can negotiate, in each country
- **Preferences**: CES within sector, Cobb–Douglas across sectors, quasi-linear outside good

$$u_n = \sum_k \beta_n^k \ln\left(\sum_i (c_{ni}^k)^{\frac{\theta_k}{\theta_k+1}}\right)^{\frac{\theta_k+1}{\theta_k}} + c_{n0}$$

- **Technology**: $q_i^k = a_i^k \ell_i^k$, iceberg costs $\tau_{ni}^k$; the outside good is made 1:1 from labour and freely traded $\Rightarrow$ all wages $= 1$
- **Equilibrium** (spending on $i$'s good $k$; total tradable spending $= 1$):

$$s_{ni}^k \equiv p_{ni}^k c_{ni}^k = \beta_n^k \frac{(p_{ni}^k)^{-\theta_k}}{\sum_j (p_{nj}^k)^{-\theta_k}}, \qquad p_{ni}^k = \frac{\tau_{ni}^k}{a_i^k}$$

- $\theta_k$ is the **trade elasticity**: low $\theta_k$ = hard to switch suppliers (semiconductors, rare earths); $\theta_k + 1$ = elasticity of substitution
- the outside good settles trade imbalances

▸ T1: derivation

# Timing: disputes and hold-up




Disputes are off the equilibrium path: prices and quantities are deterministic; only transfers depend on the threat.

- Import orders are **placed before** a dispute can arise: ex post the target cannot quickly switch to other suppliers $\Rightarrow$ **hold-up**
- The coercer threatens a comprehensive export stop; the target can threaten to retaliate
- $\lambda_{ni}$ (undirected) is the model's **geopolitical alignment**: low for allies, high for adversaries

# Step 1 — What does an export stop cost the target?

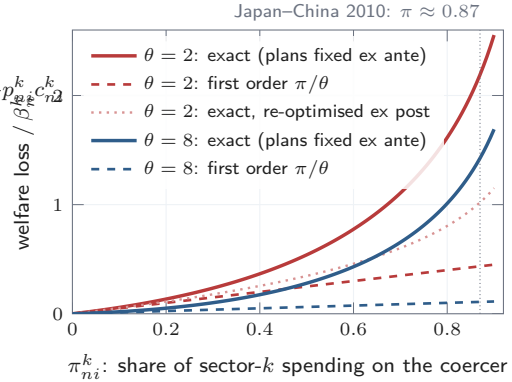
Coercer $i$ stops *all* exports to target $n$; every other order is delivered as planned (eq. 6):

$$\delta^k_{ni} = \beta^k_n \tfrac{\theta_k+1}{\theta_k}\left[\ln\sum_{j\neq i}(c^k_{nj})^{\frac{\theta_k}{\theta_k+1}} - \ln\sum_{j}(c^k_{nj})^{\frac{\theta_k}{\theta_k+1}}\right] + p^k_{ni}c^k_{ni}$$

$$\boxed{\delta^k_{ni} \;\approx\; -\frac{s^k_{ni}}{\theta_k}} \qquad (\ln(1-x)\approx -x)$$

- loss rises with **dependence** $s^k_{ni}$…
- …and with **low substitutability** (small $\theta_k$)
- the **exporter** loses (almost) nothing: its labour moves to the outside good (App. A.1: only second order)

▸ T2: exact vs. approx.




Self-made: exact loss implied by eq. (6) vs. its first-order approximation. The gap is second order, but large at extreme dependence.

# Step 2 — Bargaining in the shadow of a trade war ⇒ power

Write losses as $L_{ni} \equiv -\delta_{ni} \approx \sum_k s_{ni}^k/\theta_k$. Agreement: $n$ pays $x$ to $i$; disagreement: mutual cut-off.

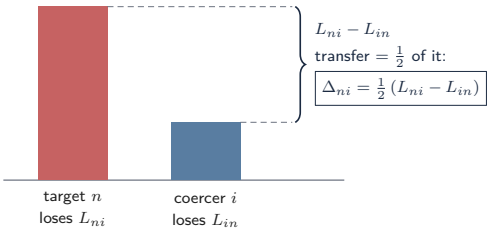
$$\max_x \; (L_{ni} - x)\,(L_{in} + x) \;\Rightarrow\; x^* = \tfrac{1}{2}(L_{ni} - L_{in})$$

$$\boxed{\Delta_{ni} \equiv \tfrac{1}{2}(L_{ni} - L_{in}) \approx \tfrac{1}{2} \sum_k \frac{1}{\theta_k}\left(s_{ni}^k - s_{in}^k\right)} \qquad (7)$$

**Power of coercer $i$ over target $n$:**

- antisymmetric: $\Delta_{ni} + \Delta_{in} = 0$
- *ex ante*: built from planned flows — a credible threat that need not be executed
- big when $n$ needs $i$ more than $i$ needs $n$, *especially in low-$\theta$ goods*: Hirschman's two sources in one number

▸ T3: Nash solution




Disagreement point = mutual trade war; symmetric Nash bargaining splits the avoided loss equally.

# Step 3 — Who negotiates, and what should governments do ex ante?

**Negotiation cost $\kappa$.** When a dispute arises (prob. $\lambda_{ni}$), the stronger side pays $\kappa$ to open talks iff $|\Delta_{ni}| \geq \kappa$:

$$\Lambda_{ni} \equiv \lambda_{ni} \cdot \mathbf{1}\{|\Delta_{ni}| \geq \kappa\} \quad \text{(prob. of negotiation)}$$

**Ex-ante welfare** (eq. 9, using $\Delta$):

$$W_n = \sum_k \beta_n^k \ln C_n^k - \sum_{k,i} \left(1 + \tfrac{\Lambda_{ni}}{2\theta_k}\right) p_{ni}^k c_{ni}^k + \sum_{k,i} \tfrac{\Lambda_{ni}}{2\theta_k}\, p_n^k q_{in}^k + \ell_n - \sum_i \Lambda_{ni} \kappa\, \mathbf{1}\{\Delta_{in} \geq \kappa\}$$

## Policy wedges

- importing from $i$ costs society *more* than its price: wedge $\Lambda_{ni}/(2\theta_k)$
- exporting to $i$ is worth *more*: wedge $\Lambda_{ni}/(2\theta_k)$
- optimal: import tariff $\eta_{ni}^k = \dfrac{\Lambda_{ni}}{2\theta_k}$, export subsidy $t_{ni}^k = \dfrac{\Lambda_{ni}}{2\theta_k}$

Private agents ignore power ⇒ they over-import from and under-export to *adversaries* (high $\lambda$), especially in low-$\theta$ goods. ▸ T4: derivation

# Two predictions, and the mapping to data

**P1 (consequences).** Shocks to preferences or technology ($\beta_n^k$, $a_i^k$) that raise $|\Delta_{ni}|$ raise $\Lambda_{ni} = \lambda_{ni}\mathbf{1}\{|\Delta_{ni}| \geq \kappa\}$: **more engagement and negotiation**.

**P2 (causes).** Shocks to relations ($\lambda_{ni}$) prompt policies that **build bilateral power**.

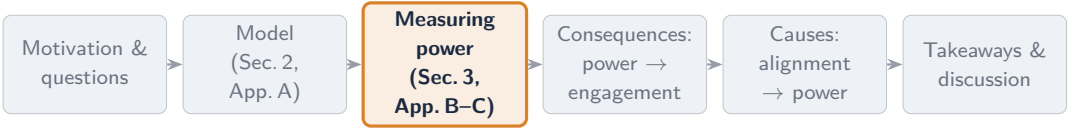
Caveat: both countries respond, in opposite directions $\Rightarrow$ net effect ambiguous. Identify *one* side's response, holding the other fixed.

| **Model** | **Empirical counterpart** |
|---|---|
| $\Delta_{ni}$ | $\text{Power}_{i\to n,t}$; $\lvert\cdot\rvert$ when undirected |
| $\Lambda_{ni}$ | ICEWS engagement index |
| $\lambda_{ni}$ | $-$Alignment (Gallup + Polity) |
| shocks to $\beta, a$ | shift-share IV: global export shifts $\times$ 1995–99 import mix |
| shocks to $\lambda$ | close-election *turnover* in partner A |
| one-sided response | B's directed power over A |

# Is $\triangle$ an artefact of the modelling shortcuts? (Appendix A + Sec. 2 remarks)

| Shortcut | Relaxation | Result |
|---|---|---|
| exporter loses nothing | A.1: irreversible ex-ante investment (Cobb–Douglas, capital share $1-\alpha$ committed) | exporter loss $\approx pq\,\frac{1-\alpha}{2\alpha}\gamma^2$: **second order** vs. importer's first order ▸ T5 |
| partial equilibrium, fixed wages | A.2: multi-sector Ricardian GE (Costinot–Donaldson–Komunjer); cut all $a$–$b$ trade by exact hat algebra | GE differential loss vs. baseline power: **corr.** $= 0.96$ ▸ T6 |
| threat covers all sectors | coercer picks one sector; type-I EV political shocks | isomorphic up to a scale constant ▸ T7 |
| same tradable spending everywhere | concession cost $\propto$ size (shares) vs. fixed (levels) | both versions implemented |
| sourcing fixed ex ante | re-optimise after sanctions | identical to first order |
| export stops, not import bans | import bans hurt the banning country | "importer power" used as a test |

# Part 3 — Measuring international power




- From $\Delta_{ni}$ to data: BACI + trade elasticities, 21 HS sections
- Eight alternative measures; who has power over whom
- Power is *not* GDP size, trade volume, or trade imbalance; why China's power rose (App. B)

# From $\Delta_{ni}$ to data

$$\bar{s}_{nit} = \sum_{k \in \mathcal{K}} \frac{1}{\theta_k} \frac{V_{nikt}}{\sum_{i'} \sum_{k'} V_{ni'k't}} \quad (10)$$

$$\boxed{\text{Power}_{i \to n,t} = \bar{s}_{nit} - \bar{s}_{int}} \quad (11)$$

- $V_{nikt}$: imports of $n$ from $i$ in sector $k$, year $t$ — **BACI** (CEPII; reconciled Comtrade, HS-6, ∼200 countries, 1995–2021)
- sample **2001–2021**: BACI lowered its recording threshold in 2000
- $\mathcal{K}$ = **21 HS sections** (ores and salt/cement moved to base metals; arms from SIPRI)
- $\theta_k$: **Broda–Weinstein (2006)** elasticities (HTS) → averaged to HS-6 → trade-weighted to sections

- one departure: the denominator is $n$'s *total imports*, not total expenditure (domestic spending by sector is rarely observed); an Eora-based variant fixes it
- variation comes from the **sectoral composition** of trade: e.g. Taiwan (semiconductors); Brazil's dependence on hard-to-replace Chinese inputs

**Takeaway.** Power is a weighted difference of two import shares: what $n$ buys from $i$ minus what $i$ buys from $n$, with hard-to-substitute sectors counting more.

# The measure is stable across eight alternative definitions

| | Correlation with baseline | |
|---|---|---|
| Alternative (Table 1, Panel A) | no FE | pair & year FE |
| Equally weighted sectors (no $\theta$) | .978 | .924 |
| Max-disruption sector (volume) | .863 | .805 |
| Max-disruption sector ($\theta$-weighted) | .761 | .677 |
| HS 2-digit sectors | .983 | .925 |
| Log trade *levels* (size matters) | .204 | .117 |
| Only sectors actually traded | .890 | .569 |
| Domestic expenditure added (Eora) | .806 | .643 |
| Trade in services added (Eora) | .924 | .846 |

- directed measure: mean $= 0$ by construction (antisymmetry); overall SD $= 0.001$
- among positive pairs: mean $0.003$, SD $0.003$
- the **levels** version is a different concept (relevant if concession costs do not scale with size)

▸ T27: Eora imputation

**Takeaway.** Choices about weights, granularity, services and domestic spending barely move the measure; the results later are robust to all of them.

# Who holds power over whom? The United States

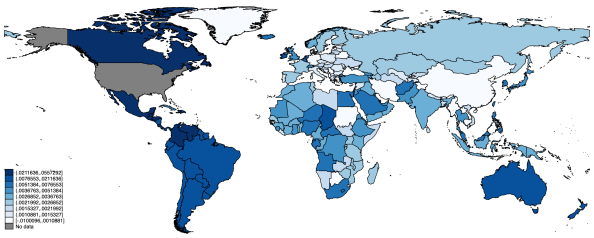



Figure 1, Panel A(a): average directed power of the US, 2001–2021 (grey = US).

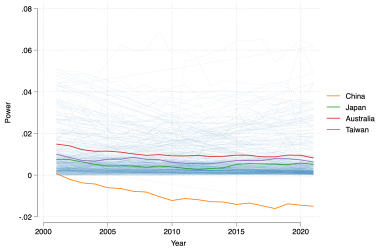



Figure 1, Panel A(b).

- Large over the Americas and US allies (Japan, Australia); small toward China despite enormous trade
- US power over China **declined**, especially 2000–2005, and turned negative

# . . . and China

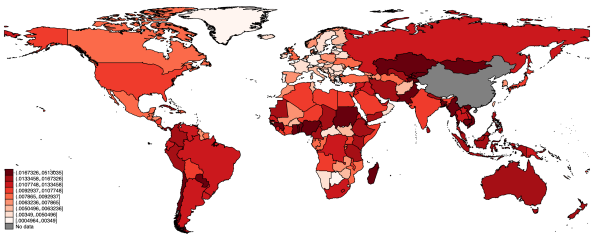



Figure 1, Panel B(a): average directed power of China, 2001–2021.

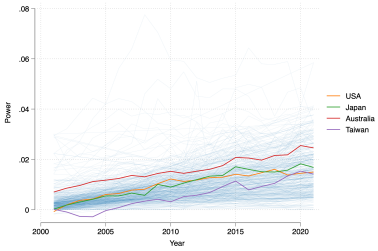



Figure 1, Panel B(b).

- China's power **rose against almost every country** after WTO entry: neighbours (India, Vietnam), East Africa, South America
- US and Chinese power over the same country are substitutes: regressing one on the other gives $\approx -0.14$ ($p < 0.001$)

# Power is not size, not volume, not the trade balance

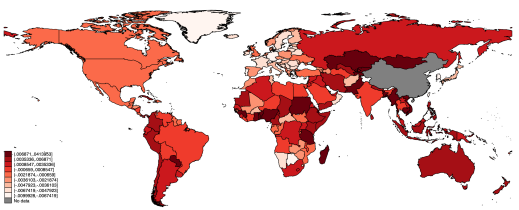


Figure A.1, Panel B: China's power after residualising GDP gap and total trade.

- **China** over-performs its GDP: positive power over the US (2001–20) and Japan (2001–10) despite a smaller GDP
- **US** "punches below its weight" — except in the Americas (Fig. A.3)

| Correlation with power | none | pair+year FE |
|---|---|---|
| GDP gap | .453 | .273 |
| GDP-per-capita gap | .135 | −.013 |
| Military-spending gap | .360 | .192 |
| Foreign-aid dependence | .094 | .038 |
| Sovereign-debt exposure | .335 | .193 |
| FBIC political bandwidth | .290 | .020 |

Table 1, Panels B–C.

- weakly related to other sources of influence — related but distinct
- most pair-years have near-zero imbalance, yet power varies a lot (Fig. A.2)

# Why did China's power rise? (Appendix B)

## B.1 Export expansion × exposure

$$\text{ChinaExposure}_{n,t} = \sum_k \text{CGMS}_{k,t} \times w_{nk\tau}$$

CGMS = China's global market share in sector $k$; $w_{nk\tau}$ = sector $k$'s share of $n$'s imports in 1995–99.

$\text{Power}_{CHN \to n,t}$ on exposure (pair & year FE): **0.491**** (0.192); $+1$ SD exposure $\approx +21.9\%$ power (median).

## B.2 Industrial policy: 10th Five-Year Plan

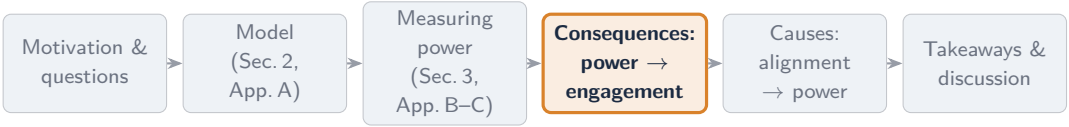
Sectors targeted in 2001–05 (e.g. nuclear reactors, furnaces, boilers; apparel), weighted by each importer's HS-4 mix:

exports $+25.7\%$ and China's power $+32.1\%$ in targeted sectors (pre: 2001–02; post: 2005–06).

*Not* evidence that the plan was geopolitically motivated.

**Takeaway.** "Exporter's global expansion × importer's pre-existing exposure" explains power shifts — exactly the logic of the shift-share instrument in Section 4.

# Part 4 — Consequences: does power induce engagement?




- Measuring engagement with ICEWS; a first look
- Strategy #1: within-pair changes (TWFE). Strategy #2: shift-share IV
- Validity, robustness, UNGA behaviour, and heterogeneity (pairs, sectors, exporter vs. importer power)

# Measuring engagement: ICEWS

- Integrated Crisis Early Warning System: machine-coded news; CAMEO event types with intensity from $-10$ (conflict) to $+10$ (cooperation) ▸ T11
- **6,733,036** bilateral events, **23,516** pairs, 2001–2021; $\approx 14$ per pair-year; 42.96% of pairs never engage; top decile $\approx 145$ per year
- mostly *not* about trade: appeals for cooperation, mediation, ministerial talks, requests to meet

**Construction:** (1) undirected; (2) keep non-violent intensities $[-7, 8)$ (92% of events); (3) count by pair $\times$ intensity bin $\times$ year, standardise *within pair*, sum the $z$-scores.

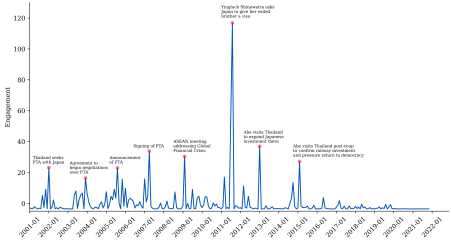



Figure A.4, Panel A: Japan–Thailand events with hand-labelled peaks (FTA talks 2002–03; 2011 visa request for Thaksin).

# A first look: engagement moves with power, not with trade volume

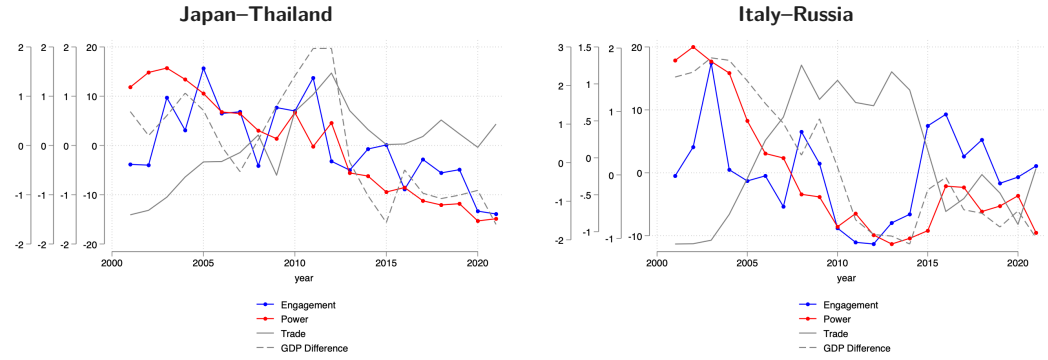



Figure A.6: all series standardised within pair and residualised on pair and year fixed effects.

- Thailand's 2011 visa request followed a rise in JPN–THA power asymmetry
- ITA–RUS engagement peaks (2006 gas cut-off; 2015 EU sanctions) coincide with high power — while total trade moves the *opposite* way

# Strategy #1: changes within country pairs

$$\text{Engagement}_{\{in\},t} = \beta_1 \text{Power}_{\{in\},t-1} + \beta_2 \text{GDPdiff}_{\{in\},t-1} + \beta_3 \text{Trade}_{\{in\},t-1} + \gamma_t + \alpha_{\{in\}} + \epsilon_{\{in\},t} \quad (12)$$

- $\text{Power}_{\{in\}} = |\text{Power}_{i \to n}|$; both directions kept, s.e. clustered by unordered pair T9
- all variables **standardised within pair** T10

| Engagement$_t$ | (1) | (2) | (3) |
|---|---|---|---|
| Power$_{t-1}$ | $0.107^{***}$ | $0.106^{***}$ | $\mathbf{0.236}^{***}$ |
| | (0.0115) | (0.0115) | (0.0163) |
| GDP difference$_{t-1}$ | | $0.0860^{***}$ | $0.0923^{***}$ |
| Total trade$_{t-1}$ | | | $-0.242^{***}$ |
| Pair FE, year FE | ✓ | ✓ | ✓ |
| $N$ | | 517,298 | |

Table 2, columns 1–3.

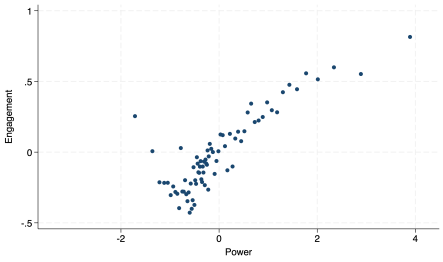



Figure 2: binscatter (`binsreg`) with controls, pair & year FE.

T12: binscatter

# How big is it?

**0.236**
index units per $+1$ SD of lagged power (col. 3)

**+38.4%**
more engagement next year, at the mean

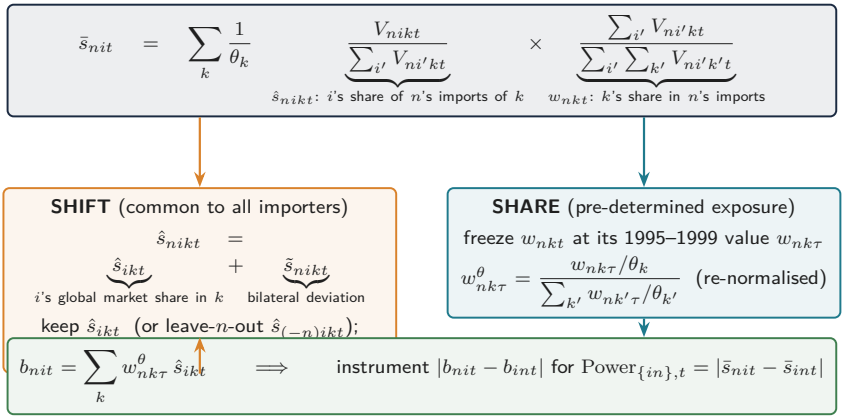
**+18.8%**
evaluated at the median pair

**−0.242**
total trade enters *negatively*: asymmetry, not volume

- Controlling for GDP gap and bilateral trade *raises* the coefficient ($0.107 \to 0.236$): volume and asymmetry pull in opposite directions
- Close to linear in standardised units (Fig. 2); raw units show mild concavity (Fig. A.7) — extreme concentration lives below the HS-section level
- Left-tail outliers: close allies with large, lopsided trade (e.g. US–Canada)

**Takeaway.** A one-SD rise in power asymmetry is followed by markedly more bilateral engagement — and not only among great powers (excluding USA, China, Russia: $0.219$).

# Strategy #2: a shift-share instrument for power




**Example** (Fig. A.8): Germany's world market share fell by similar amounts in stone/glass ($-0.513$ pp) and textiles ($-0.436$ pp), yet China's 1995–99 exposure to the two differs by $11.6$ pp $\Rightarrow$ different predicted changes in German power over China. ▸ T13

# IV results: power causes engagement

| | (4) | (5) | (6) |
|---|---|---|---|
| *First stage: Power$_{t-1}$ on $\lvert b_{nit} - b_{int} \rvert$ (Table A.4)* | | | |
| IV | $0.313^{***}$ | $0.314^{***}$ | $0.222^{***}$ |
| | (0.00366) | (0.00367) | (0.00351) |
| F-statistic | 7,312 | 7,322 | 3,996 |
| *Second stage: Engagement$_t$ (Table 2)* | | | |
| Power$_{t-1}$ | $0.454^{***}$ | $0.439^{***}$ | $\mathbf{0.697}^{***}$ |
| | (0.0412) | (0.0412) | (0.0589) |
| GDP difference | | $0.0709^{***}$ | $0.0858^{***}$ |
| Total trade | | | $-0.539^{***}$ |
| $N$ (pairs with 1995–99 trade) | | 486,026 | |

**Reading the IV**

- very strong first stage (F in the thousands)
- IV ($0.697$) $>$ OLS ($0.236$). Consistent with: measurement error in power (attenuation); a LATE for power shifts driven by *exporters' global expansion*; or OLS biased down by strategic, partner-specific trade
- the IV purges bilateral, possibly strategic trade manipulation and reverse causality from engagement to trade

# Is the shift-share IV credible? (I) Which sectors drive it?

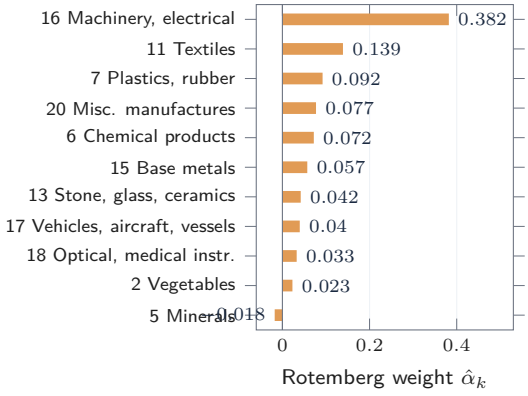



Table A.6 (top 10 + the negative one). Self-made chart.

**Rotemberg weights** (Goldsmith-Pinkham, Sorkin & Swift 2020): $\hat{\beta}_{\text{Bartik}} = \sum_k \hat{\alpha}_k \hat{\beta}_k$ — which sector-instruments the estimate leans on. ▸ T14

- machinery & electrical $= 0.38$; top five $\approx 0.76$
- **balance tests**: the top-5 sectors' 1995–99 shares show moderate, mixed-sign correlations with changes in economic, military and political conditions
- **drop the top-5 sectors**: IV $= 0.647$ (0.068) (vs. 0.697)
- **leave-one-out shifter** (Autor, Dorn & Hanson 2013; exclude $n$ from $i$'s market share): $0.703$ (0.060) ▸ T15

# Is the shift-share IV credible? (II) Many instruments and inference

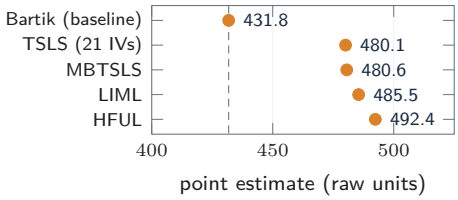



Table A.8 (raw units, directed sample). Self-made chart.

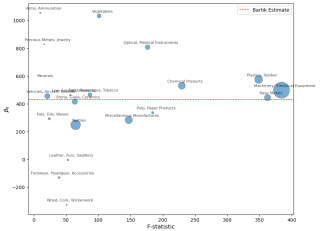


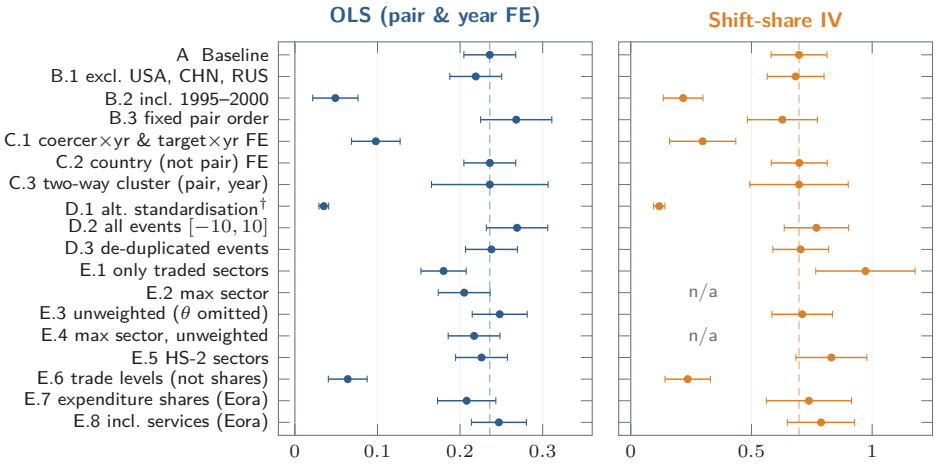

Figure A.9: just-identified $\hat{\beta}_k$ vs. first-stage F; size = Rotemberg weight.

**Use the 21 sector instruments separately**: TSLS, MBTSLS, LIML and HFUL all give $\approx 480$–$492$ vs. Bartik $431.8$ ▸ T16

- over-ID tests (Anderson–Rubin; Chao–Hausman–Newey–Swanson–Woutersen) **reject** homogeneity ▸ T17
- but sectors with large weight or high F cluster around the baseline; the dispersion comes from low-weight sectors

**Shift-share-robust inference** (Adão, Kolesár & Morales 2019), sector clusters: $431.8$ (s.e. $12.543$) vs. pair-clustered s.e. $64.744$ ▸ T18

# Robustness: 20+ variations (Table A.9)



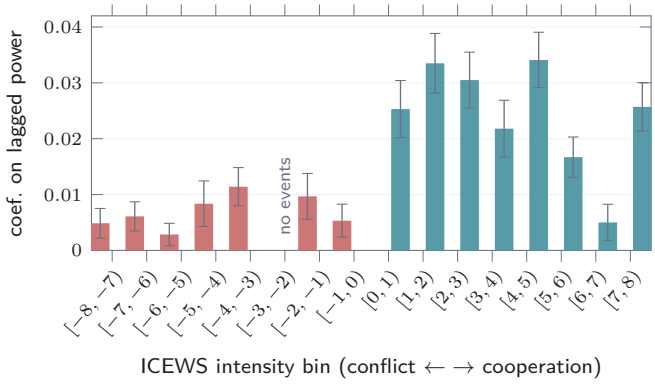

Self-made from Table A.9; 95% CIs. †D.1: total event count standardised (other scale). Raw-unit rows (C.4, F.1, F.2) omitted — all $p < 0.01$.

# What kind of engagement responds?

**By theme** (Table A.2, OLS):

| | |
|---|---|
| Sanction-related | $-0.0135$ |
| | (0.0105) |
| Economic, not sanctions | $0.0494^{***}$ |
| | (0.0078) |
| Non-economic | $0.0496^{***}$ |
| | (0.0084) |

- **not** driven by sanctions
- economic and non-economic engagement rise alike




Self-made from Table A.3 (one regression per intensity bin). Positive and significant in **every** intensity bin, from threats and demands to cooperation.

# Does power change behaviour? UNGA voting (Sec. 4.6, App. D)

**Why UNGA:** negotiation outcomes are rarely public, but all members vote on the *same* resolutions ⇒ comparable positions.

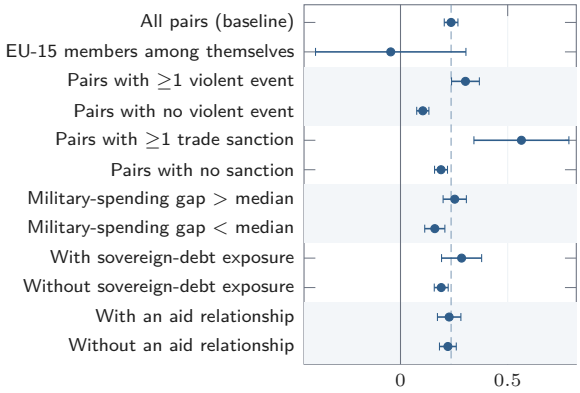
- positions: dynamic ordinal IRT ideal points (Bailey, Strezhnev & Voeten 2017) ▸ T21
- outcome: pairwise vote similarity; control for its lag

| UNGA similarity$_t$ (Table A.10) | OLS (3) | IV (6) |
|---|---|---|
| Power$_{t-1}$ | $0.0105^{***}$ | $0.0350^{***}$ |
| | (0.00263) | (0.0106) |
| Similarity$_{t-1}$ | $0.450^{***}$ | $0.450^{***}$ |
| Controls, pair & year FE | ✓ | ✓ |
| $N$ | 243,450 | 229,574 |

More power asymmetry ⇒ the pair **votes more alike** afterwards, relative to its own history.

Consistent with accommodation or concessions — not direct proof (Schelling: whether a move is a threat or a promise depends on the baseline).

# Where does power bite? Institutions, conflict, other influence




- **EU-15 among themselves**: no effect (IV $-0.22$, s.e. $0.46$); random EU–non-EU samples centre near $0.2$ (Fig. A.10)
- **pairs with past violence or sanctions**: stronger (IV $0.65$ vs. $0.52$; $1.07$ vs. $0.62$) — realised conflict makes threats credible
- **military / debt asymmetry**: stronger; controlling for them leaves power unchanged ($0.244$, $0.227$) — *complements*

Self-made from Tables A.11, A.12, A.16 (OLS, col. 3).

# Which sectors generate power? Low elasticity, complex, critical

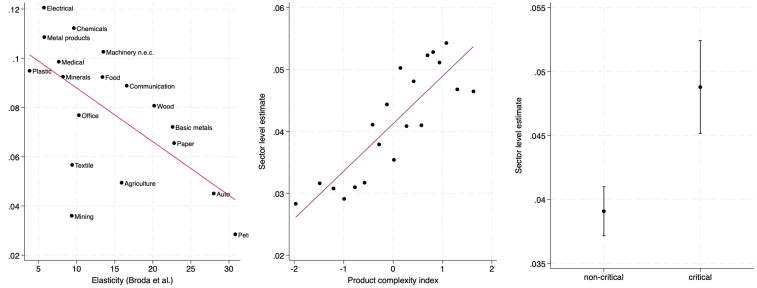
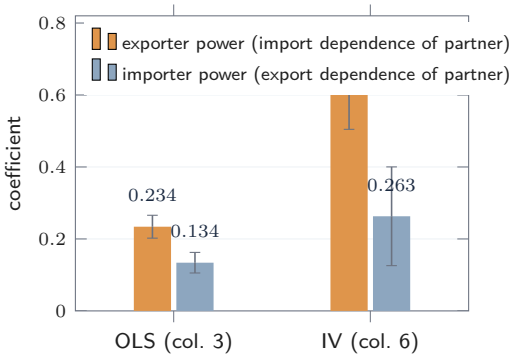



Figure 3: sector-specific power–engagement coefficients vs. trade elasticity (CP sectors), product complexity (HS-4), critical goods (HS-4).

- recompute power sector by sector (HS-4, *unweighted*) and re-run eq. (12) ▸ T19
- model: without $1/\theta$ weights, coefficients should be **larger in low-$\theta$ sectors** — they are
- slope on elasticity: $-0.0022^{***}$ (BW), $-0.0013^{***}$ (CP)
- PCI $+0.008^{***}$; critical $+0.010^{***}$ (HS-4); standouts: uranium ore, transformers, optical fibres, semiconductors

# Exporter power vs. importer power



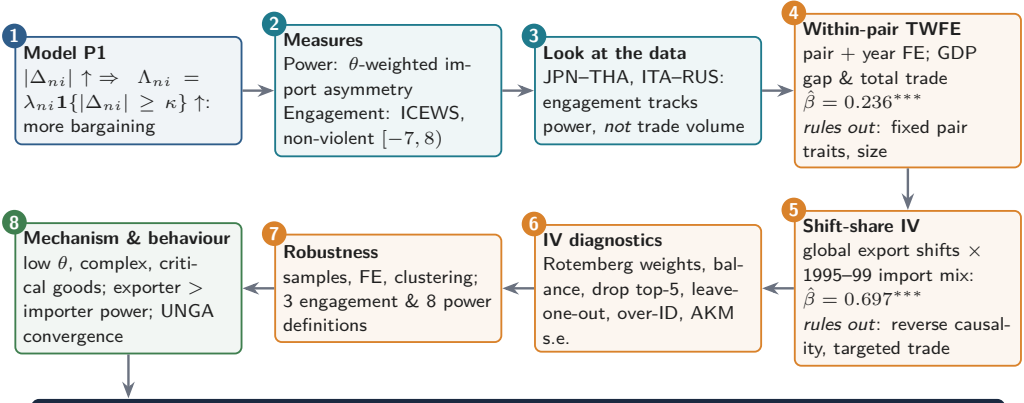

Self-made from Table A.15 (both measures in one regression).

**Prediction** (model + App. A.1): disruption hurts the *importer* to first order but the exporter only to second order. So power from the partner's **import** dependence (baseline, "exporter power") should matter more than power from its **export** dependence ("importer power"):

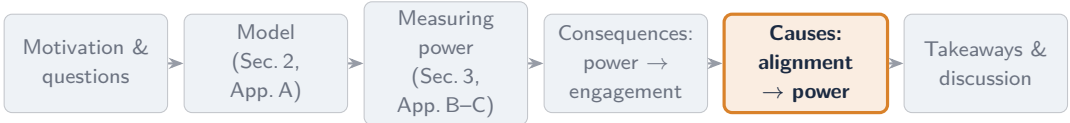
$$\text{Power}^{\text{imp}}_{i\to n,t} = \sum_k \frac{1}{\theta_k}\left(\frac{V_{inkt}}{\sum_{k'}\sum_j V_{jnk't}} - \frac{V_{nikt}}{\sum_{k'}\sum_j V_{jik't}}\right)$$

**Result**: importer power is positive but about **half** as large, and only sometimes significant.

# The path to Conclusion 1

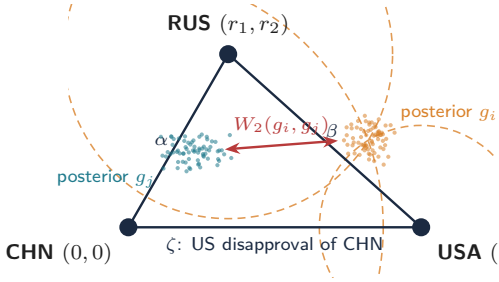


# Part 5 — Causes: do adversarial relations prompt power build-up?




- A time-varying, trade-free measure of geopolitical alignment (App. F)
- Close-election turnover in A as a shock to the A–B relationship; B's response
- First stage, checks, main estimate, robustness, and *how* power is built

# Measuring alignment without using trade




Self-made illustration of "statistical multilateration".

Needs: cross-section *and* time variation; all pairs; **no trade or policy inputs** (so not UNGA).

1. **Gallup World Poll**: $\approx 1{,}000$ people $\times$ 164 countries a year rate US, Russian, Chinese leadership $\Rightarrow$ disapproval = "distance" to 3 anchors
2. anchors: CHN $(0,0)$, USA $(\zeta, 0)$, RUS by triangle geometry
3. Gaussian likelihood over a 2-D plane $\rightarrow$ posterior ▸ T22
4. 100k rejection draws $\rightarrow$ $k$-means to 1,000 weighted points ▸ T23
5. $d_{ij}^{\text{Gallup}}$ = **Wasserstein-2** distance of posteriors

$$\text{Alignment}_{ijt} = 1 - \left[\alpha\, d_{ijt}^{\text{Gallup}} + (1-\alpha)\, d_{ijt}^{\text{Polity}}\right], \quad \alpha = 0.5$$

# Does the alignment measure make sense?

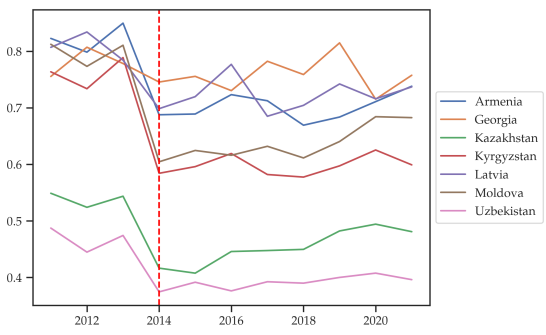



Figure A.15: former Soviet states' alignment with Ukraine; dashed line = 2014 annexation of Crimea.

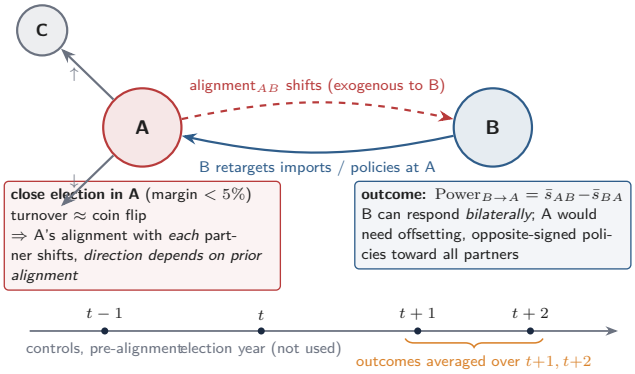
**Validation 1:** a sharp drop with Ukraine after 2014 — although Ukraine's bilateral ties are never observed directly, only inferred.

**Validation 2:** predicts UNGA vote similarity (Table A.17):

| FE | none | pair | year | both |
|---|---|---|---|---|
| coef. | $.064^{***}$ | $.013^{***}$ | $.065^{***}$ | $.017^{***}$ |

The correlation is mostly cross-sectional; the new measure has $\geq 6\times$ more over-time variation than UNGA-based alignment (ANOVA).

# Design: close elections in A, response of B




**Challenge 1**: alignment may itself respond to trade $\Rightarrow$ use shocks from A's *domestic* close elections.

**Challenge 2**: both sides respond $\Rightarrow$ outcome is B's *directed* power over A; A's turnover moves its alignment with different partners in *opposite* directions.

**Data**: Marx, Pons & Rollet (2024), 177 countries, 2012–18; close = margin $< 5\%$ $\Rightarrow$ 33 close elections, 13 turnovers (Table A.18). ▸ T25

# Examples: the US in 2016 and Poland in 2015

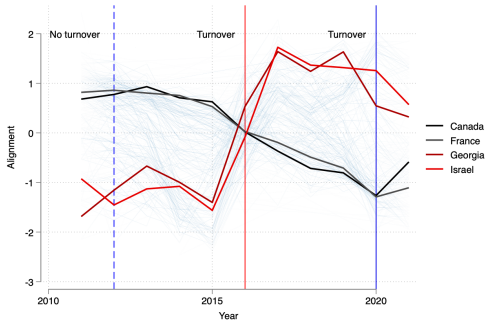



Fig. A.16 A: alignment with the US (illustration only — 2016 was not "close": 304–227 electoral votes).

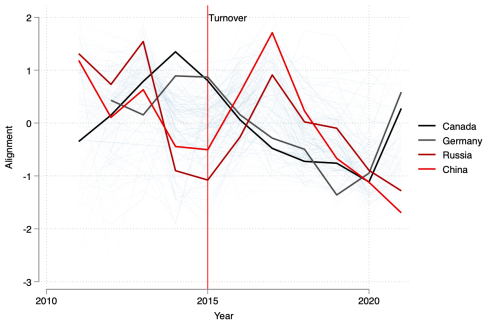



Fig. A.16 B: alignment with Poland around its 2015 close election (in the sample).

Turnover *reverses* trajectories: Canada and France (Germany, for Poland) turn away, Georgia and Israel (Russia and China, for Poland) move closer. No shift after the 2012 US election (no turnover).

# First stage: turnover reverses the persistence of alignment

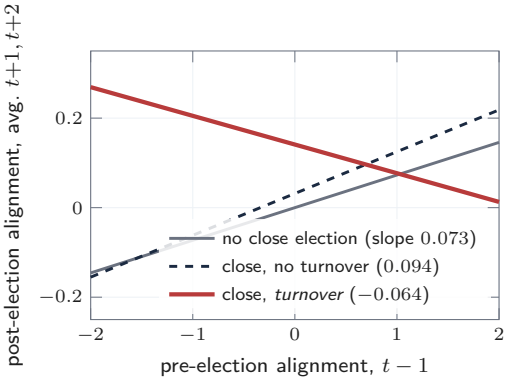
$$a_{\{t+1,t+2\}} = \beta_1 a_{t-1} + \beta_2 \mathrm{TO}_t + \beta_3\, a_{t-1}\mathrm{TO}_t + \beta_4 \mathrm{NT}_t + \beta_5\, a_{t-1}\mathrm{NT}_t + \text{controls}_{t-1} + \alpha_{\{AB\}} + \gamma_t \qquad (18)$$

$a$ = alignment of pair AB; TO / NT = close election in A with / without turnover in year $t$.

Table 3A, col. (3)

| | |
|---|---|
| Alignment$_{t-1}$ | $0.0729^{***}$ (0.00405) |
| Turnover (TO) | $0.141^{***}$ (0.0243) |
| Alignment$_{t-1}\times$TO | $-0.137^{***}$ (0.0251) |
| Close, no turnover (NT) | $0.0316^{*}$ (0.0179) |
| Alignment$_{t-1}\times$NT | $0.0205$ (0.0186) |

After a turnover the slope on past alignment flips sign ($0.073 - 0.137 = -0.064$): **trend reversal** — former rivals converge, former friends drift apart.




Self-made: implied lines from Table 3A, other covariates at zero. Data version: Fig. A.17.

# Threats to identification — and the checks

| Threat | Check | Result |
|---|---|---|
| Turnover anticipated, or selected on pair conditions | regress turnover on B's and pair-level economic, international, military and political conditions at $t-1$; FDR-adjusted (Benjamini–Krieger–Yekutieli) ▸ T26 | nothing significant (Table A.19); $\text{Alignment}_{t-1}$ also absorbs known trends |
| Exclusion: A's turnover changes trade with *everyone* | effect of turnover on the *average* alignment / power of A's partners | alignment $0.019$ (0.027), power $0.025$ (0.023): no average effect (Table A.20) |
| Exclusion: A retargets its policy at B | A's targeted trade policies toward B (Global Trade Alert) | none significant (Table A.25) ▸ T28 |
| Arbitrary A/B labels for pairs without elections | 50 random designations | all $\approx$ baseline (Fig. A.18) |

# Main result: worse alignment ⇒ B builds power over A

$$\text{Power}_{B\to A,\{t+1,t+2\}} = \beta_1\,\hat{a}_{\{AB\},\{t+1,t+2\}} + \beta_2\,a_{\{AB\},t-1} + \text{controls}_{t-1} + \alpha_{\{AB\}} + \gamma_t + \epsilon \qquad (19)$$

| $\text{Power}_{B\to A}$ (Table 3, Panel B) | (1) | (2) | (3) |
|---|---|---|---|
| $\widehat{\text{Alignment}}_{\{t+1,t+2\}}$ | $-0.368^{**}$ | $-0.381^{***}$ | $\mathbf{-0.386}^{**}$ |
| | (0.145) | (0.147) | (0.147) |
| GDP difference control | | ✓ | ✓ |
| Total trade control | | | ✓ |
| Pair FE, year FE | ✓ | ✓ | ✓ |
| $N$ | | 55,819 | |

Controls at $t-1$: alignment, GDP gap, total trade, $\text{Power}_{B\to A}$.

**+0.39 SD**

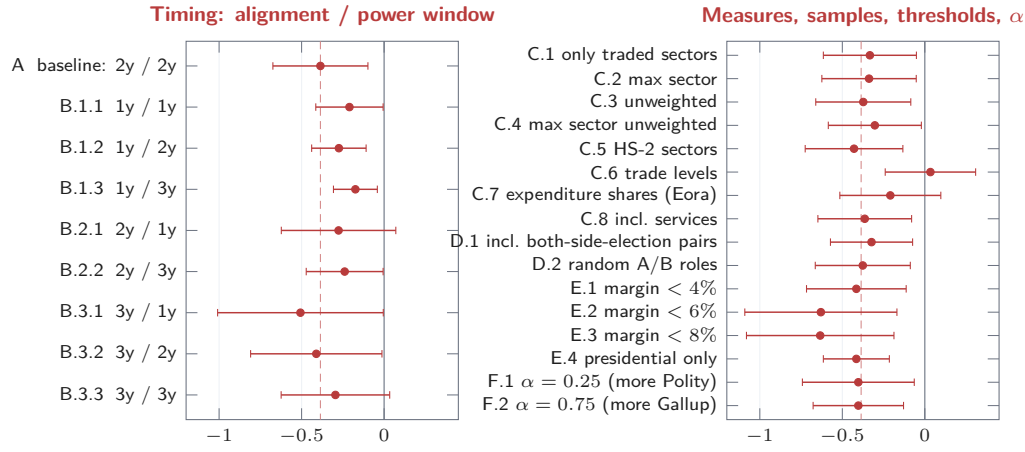
power of B over A per $-1$ SD alignment

**50th → 85th**

percentile of power, for a pair starting at the median

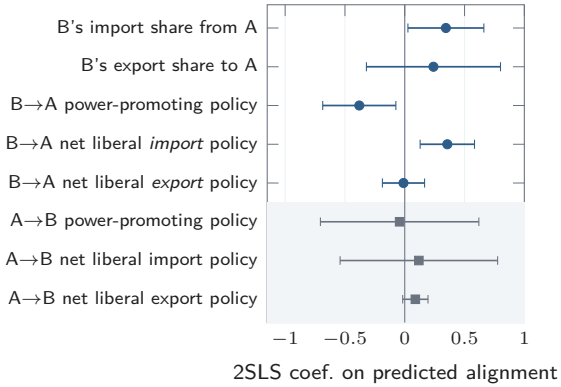
**Takeaway.** Trade is (partly) strategic: countries use trade dependence to manage the risks of shifting alignments.

# Robustness of the build-up result (Table A.21)




Self-made from Table A.21, col. 3; 95% CIs. Exceptions: power in trade *levels* (C.6, null — build-up seems to scale with size) and Eora expenditure shares (C.7, same sign, insignificant).

# How is power built? Cutting imports, via import policy



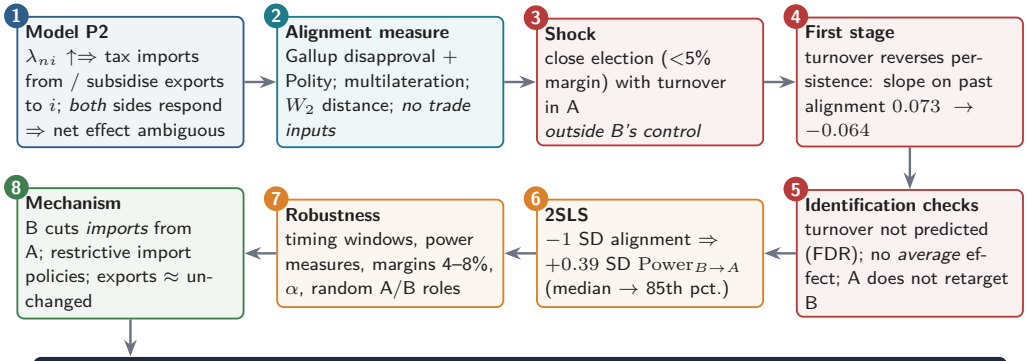

Self-made from Table 4 and Table A.25 (shaded rows: the electing country A, a placebo).

**Sign reading**: positive = "lower alignment ⇒ less of it".

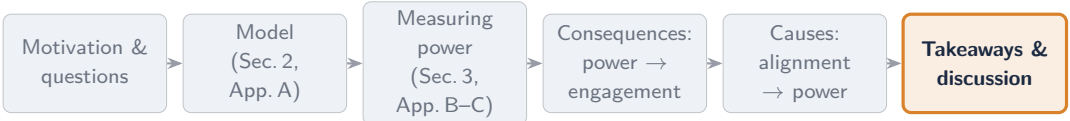
- B's **import share** from A falls (0.344**); its export share barely moves (0.240, n.s.)
- B adopts **power-promoting** policies (-0.380**): fewer liberal / more restrictive *import* measures (0.356***); export policy flat
- A shows **no** targeted response

**Defensive**, not offensive: diversifying sourcing is cheaper than quickly ramping up exports.

# The path to Conclusion 2



# Part 6 — Takeaways and discussion




- What we learned; implications for trade policy
- What the paper does not do
- Questions for discussion

# Conclusions

1. **Trade creates power** that is distinct from economic size, trade volume and imbalances: it lives in the *sectoral composition* of dependence, especially in hard-to-substitute goods.
2. **Power has consequences**: more asymmetry $\Rightarrow$ more bilateral engagement and negotiation, and convergence in UN votes (consistent with concessions).
3. **Power is built strategically**: when relations sour, countries raise their power over the adversary — mostly *defensively*, by cutting import dependence through import policy.

## Implications

- **Bilateral imbalances can matter** geopolitically, even if multilateral balances are macro outcomes — a precise version of Viner's (1948) "power and plenty".
- **Efficiency vs. security**: comparative-advantage trade is positive-sum; accumulating power to extract rents is negative-sum and sacrifices efficiency at home and abroad.

# What the paper does not (yet) do

- **Direct links only**: no networks, third countries or supply-chain (indirect) dependence
- **Countries as units**: blocs change the picture — each EU-15 member has *negative* power over China, the EU as a bloc *positive* (Fig. A.19)
- **Outcomes of negotiations** are unobserved (no counterfactual); UNGA convergence is suggestive
- **Static model**: disputes off-path; no dynamics of accumulation or retaliation
- **Interaction** with military, aid and debt power is correlational

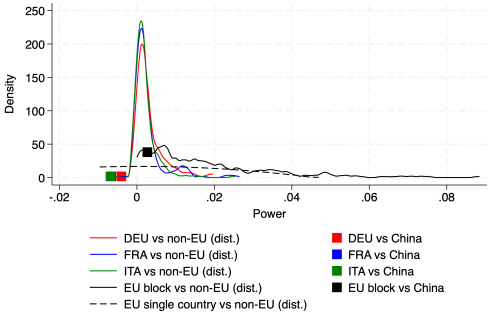



Figure A.19: EU members vs. the EU bloc, power over non-EU countries (2010).

# Questions for discussion

1. **Elasticities.** $\theta_k$ comes from US import data (Broda–Weinstein) and is applied to every importer; the model's $\theta$ is $\sigma - 1$. How sensitive are the weights? (The unweighted variant is reassuring.)
2. **Aggregation.** HS sections average away choke points (the 87% rare-earth dependence is HS-6); the first-order loss understates extreme dependence (Step 1 chart).
3. **Outcome scale.** The engagement index is a *sum* of category $z$-scores, so "SD" language and the 38%/19% conversions depend on it (D.1: $0.035$ when the total is standardised).
4. **Shift-share.** 21 sectors; machinery carries 38% of the weight; over-ID rejects homogeneity; the AKM s.e. $(12.5)$ is *smaller* than the pair-clustered one $(64.7)$ — unusual with only 21 shock clusters.
5. **Close elections.** 13 turnovers drive the first stage; the shocks are election-level, but s.e. are pair-clustered. "No-turnover" close elections also enter the first stage.
6. **Interpretation.** More engagement is consistent with coercion *and* with cooperation; what share is bargaining over concessions?

# Thank you!

## Questions & discussion

**Backup: technical toolkit (T1–T28) and additional results**

# Backup I

## Technical toolkit (T1–T28)

# Technical toolkit: index (click to jump)

**Model (Sec. 2, App. A)**

T1 CES demand and shares
T2 Welfare loss of a cut-off
T3 Nash bargaining $\Rightarrow \Delta$
T4 Ex-ante wedges, optimal policy
T5 Irreversible investment
T6 CDK GE and exact hat algebra
T7 Targeted sanctions (Gumbel)

**Measurement**

T8 Trade elasticities
T11 ICEWS engagement index
T27 Eora imputations
T28 Global Trade Alert coding

**Strategy 1–2: power $\rightarrow$ engagement**

T9 TWFE, HDFE, dyadic clustering
T10 Within-pair standardisation
T12 Binscatter with controls
T13 Shift-share IV: identification
T14 Rotemberg weights
T15 Balance tests, leave-one-out
T16 Many-IV estimators (k-class, HFUL)
T17 Over-ID and weak-IV tests
T18 AKM shift-share s.e.
T19 Sector regressions, PCI, critical
T20 Placebo by random draws
T21 UNGA ideal points (IRT)

**Strategy 3: alignment $\rightarrow$ power**

T22 Statistical multilateration
T23 Rejection sampling, $k$-means
T24 Wasserstein / optimal transport
T25 Close-election IV
T26 FDR: BH and BKY q-values

Each slide: **why** it is needed $\rightarrow$ **formula / derivation** $\rightarrow$ **implementation** $\rightarrow$ **caveats**. Full derivations and code are in the handout, Part V.

# T1 CES demand and the expenditure-share equation

**Problem** (country $n$, wage $= 1$):

$$\max_{\{c_{ni}^k\}} \sum_k \beta_n^k \ln C_n^k - \sum_k \sum_i p_{ni}^k c_{ni}^k, \qquad C_n^k = \left( \sum_i (c_{ni}^k)^{\rho_k} \right)^{1/\rho_k}, \quad \rho_k = \frac{\theta_k}{\theta_k + 1}$$

**Step 1 (within sector)**: cost minimisation gives

$$c_{ni}^k = \left( \frac{p_{ni}^k}{P_n^k} \right)^{-(\theta_k+1)} C_n^k, \qquad P_n^k = \left( \sum_i (p_{ni}^k)^{-\theta_k} \right)^{-1/\theta_k}$$

**Step 2 (across sectors)**:
$\max_C \ \beta \ln C - PC \Rightarrow P_n^k C_n^k = \beta_n^k$ (quasi-linearity: no income effects).

**Result** (paper's eq. 5):

$$s_{ni}^k \equiv p_{ni}^k c_{ni}^k = \beta_n^k \frac{(p_{ni}^k)^{-\theta_k}}{\sum_j (p_{nj}^k)^{-\theta_k}} \equiv \beta_n^k \pi_{ni}^k$$

- $\sigma_k = \theta_k + 1$: elasticity of substitution; $\partial \ln(s_{ni}/s_{nj}) / \partial \ln(p_{ni}/p_{nj}) = -\theta_k =$ **trade elasticity**
- same share structure as gravity models (Eaton–Kortum; Anderson–van Wincoop)
- identity used in T2: $\dfrac{(c_{ni}^k)^{\rho_k}}{\sum_j (c_{nj}^k)^{\rho_k}} = \pi_{ni}^k$, since $c \propto p^{-(\theta+1)} \Rightarrow c^\rho \propto p^{-\theta}$

back to talk

# T2 Welfare loss of losing a supplier: exact vs. first order

Let $\pi \equiv \pi^k_{ni}$ (within-sector share). **Plans fixed ex ante** (eq. 6): $i$'s term drops out of the CES sum, $\sum_{j\neq i} c^\rho = (1-\pi)\sum_j c^\rho$, and unspent $\beta\pi$ goes to the outside good:

$$\frac{\delta^{\text{ante}}}{\beta} = \frac{\theta+1}{\theta}\ln(1-\pi) + \pi = -\frac{\pi}{\theta} - \frac{\theta+1}{2\theta}\pi^2 + O(\pi^3)$$

**Re-optimised ex post**: $P' = P(1-\pi)^{-1/\theta}$, indirect utility $\beta\ln(\beta/P)$:

$$\frac{\delta^{\text{post}}}{\beta} = \frac{1}{\theta}\ln(1-\pi) = -\frac{\pi}{\theta} + O(\pi^2)$$

- both $\approx -\pi/\theta$: first order equal (**envelope theorem**); gap $\delta^{\text{post}} - \delta^{\text{ante}} = \beta[-\ln(1-\pi) - \pi] = \beta[\frac{\pi^2}{2} + \ldots] \geq 0$
- ex-post version = Arkolakis–Costinot–Rodríguez-Clare type formula $\hat{W} = \hat{\lambda}^{1/\theta}$
- summing over sectors: $\delta_{ni} \approx -\sum_k s^k_{ni}/\theta_k$
- **caveat**: at $\pi = 0.87, \theta = 2$ the exact ex-ante loss is $2.19\beta$ vs. $0.44\beta$ to first order — the linear measure is designed for modest shares

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# T3 Nash bargaining ⇒ the power formula

Payoffs relative to the status quo: agreement $n : -x$, $i : +x$; threat point (mutual cut-off) $n : -L_{ni}$, $i : -L_{in}$, with $L \equiv -\delta \geq 0$.

**Generalised Nash** (weight $b$ for $n$):

$$\max_x \; (L_{ni} - x)^b \, (L_{in} + x)^{1-b}$$

FOC: $(1-b)(L_{ni} - x) = b(L_{in} + x)$

$$\Rightarrow \; x^* = (1-b)L_{ni} - bL_{in} \overset{b=1/2}{=} \tfrac{1}{2}(L_{ni} - L_{in}) \equiv \Delta_{ni}$$

- each side gains $\frac{1}{2}(L_{ni} + L_{in})$ over the threat point: avoided dead-weight loss split equally
- $\Delta_{in} = -\Delta_{ni}$ (antisymmetric); with $L \approx \sum_k s/\theta$: $\Delta_{ni} \approx \frac{1}{2} \sum_k (s^k_{ni} - s^k_{in})/\theta_k$
- reduced form: holds third countries fixed; Nash split could be micro-founded by alternating offers (Rubinstein) with equal patience
- **notation**: eq. (6) defines $\delta \leq 0$ as a welfare *change*; eq. (7) uses magnitudes — we write $L = -\delta$

◂ back to talk

# T4 Ex-ante welfare wedges and the optimal tariff / subsidy

**Expected dispute payoff** (eq. 8):
$L_n = -\sum_i \Lambda_{ni}\big(\Delta_{ni} + \kappa \mathbf{1}\{\Delta_{in} \geq \kappa\}\big)$.
Use $s^k_{ni} = p^k_{ni} c^k_{ni}$ (spending on imports) and
$s^k_{in} = p^k_{in} c^k_{in} = p^k_n q^k_{in}$ (value of $n$'s exports to $i$):

$$L_n = -\sum_{k,i} \tfrac{\Lambda_{ni}}{2\theta_k} p^k_{ni} c^k_{ni} + \sum_{k,i} \tfrac{\Lambda_{ni}}{2\theta_k} p^k_n q^k_{in} - \sum_i \Lambda_{ni} \kappa \mathbf{1}\{\cdot\}$$

Set $T_n = L_n$ in utility $\Rightarrow$ eq. (9).

**FOCs** (price-taking planner, $\Lambda$ locally fixed):

$$\frac{\partial\, \beta \ln C}{\partial c^k_{ni}} = \left(1 + \tfrac{\Lambda_{ni}}{2\theta_k}\right) p^k_{ni} \quad \text{vs. private:} \quad = p^k_{ni}$$

- ad-valorem **import tariff** $\eta^k_{ni} = \Lambda_{ni}/(2\theta_k)$ (revenue rebated) implements the planner's choice
- **export subsidy** $t^k_{ni} = \Lambda_{ni}/(2\theta_k)$ aligns private and social value of exports to $i$
- wedge $\uparrow$ in $\lambda_{ni}$ (adversaries), $\downarrow$ in $\theta_k$; zero if $|\Delta_{ni}| < \kappa$
- **caveats**: $\Lambda$ jumps at $|\Delta| = \kappa$; policy moves $\Delta$ itself; partner retaliates $\Rightarrow$ net effect of $\lambda$ shocks ambiguous (why Sec. 5 isolates one side)

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# T5 Irreversible investment: exporter losses are second order (App. A.1)

Technology $q = a\left(\frac{\iota}{1-\alpha}\right)^{1-\alpha}\left(\frac{x}{\alpha}\right)^{\alpha}$; ex ante $\iota = (1-\alpha)pq$ is sunk, $p = 1/a$.
Ex post, to produce $\tilde{q}$: $\tilde{x} = \alpha\,\frac{q}{a}\,(\tilde{q}/q)^{1/\alpha}$.

Losing share $\gamma$ of output ($\tilde{q} = (1-\gamma)q$), the loss of quasi-rent $p\tilde{q} - \tilde{x}$ is

$$\text{Loss} = pq\left[\alpha(1-\gamma)^{1/\alpha} - \alpha + \gamma\right]$$

Expand $(1-\gamma)^{1/\alpha} = 1 - \frac{\gamma}{\alpha} + \frac{1}{\alpha}\left(\frac{1}{\alpha} - 1\right)\frac{\gamma^2}{2} + O(\gamma^3)$:

$$\boxed{\text{Loss} \approx pq\,\frac{1-\alpha}{2\alpha}\,\gamma^2}$$

- first-order terms cancel: **envelope theorem** — competitive firms earn no pure rent, and the flexible input re-optimises
- importer's loss $\approx s/\theta$ is **first order** $\Rightarrow$ asymmetric threat, so power = *import* dependence
- explicit coefficient $\frac{1-\alpha}{2\alpha}$ is our derivation (the paper shows the first-order terms vanish)
- breaks with market power/markups or endogenous factor prices $\Rightarrow$ paper also tests "importer power"

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# T6 GE check: multi-sector Ricardian model and exact hat algebra (App. A.2)

**CDK model** (Fréchet productivity, Cobb–Douglas sectors):

$$S_{ni}^k = \frac{(w_i \tau_{ni}^k / z_i^k)^{-\theta_k}}{\sum_j (w_j \tau_{nj}^k / z_j^k)^{-\theta_k}}, \qquad w_i \ell_i = \sum_{n,k} \beta_n^k w_n \ell_n S_{ni}^k$$

**Hat algebra** ($\hat{x} = x'/x$, $\hat{z} = 1$):

$$\hat{S}_{ni}^k = \frac{(\hat{w}_i \hat{\tau}_{ni}^k)^{-\theta_k}}{\sum_j S_{nj}^k (\hat{w}_j \hat{\tau}_{nj}^k)^{-\theta_k}}, \quad \hat{w}_i w_i \ell_i = \sum_{n,k} \beta_n^k \hat{w}_n w_n \ell_n S_{ni}^k \hat{S}_{ni}$$

$$\hat{P}_n^k = \Big[ \sum_i S_{ni}^k (\hat{w}_i \hat{\tau}_{ni}^k)^{-\theta_k} \Big]^{-1/\theta_k}, \quad \hat{W}_n = \hat{w}_n \Big/ \prod_k (\hat{P}_n^k)^{\beta_n^k}$$

**Counterfactual**: $(\hat{\tau}_{ab}^k)^{-\theta} = (\hat{\tau}_{ba}^k)^{-\theta} = 0 \ \forall k$. GE power analogue: $\Delta_a - \Delta_b = \hat{W}_b - \hat{W}_a$, $\Delta_c = 1 - \hat{W}_c$.

**Algorithm**: $\hat{w} \leftarrow 1$; repeat {compute $\hat{S}$; update $\hat{w}$ from trade balance; normalise $\hat{w}_{i_0} = 1$; damp} until $\|\Delta \hat{w}\| < 10^{-10}$.

- needs only observed shares $S$, $\beta$, incomes, $\theta$ — no productivity or trade-cost *levels*
- result: **corr.** $0.96$ with the baseline measure — terms-of-trade effects are small

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# T7 Targeted sanctions with type-I extreme-value shocks

Coercer cuts *one* sector $k$; damage $\delta_k = s^k_{ni}/\theta_k$ plus a political shock $\sigma\varepsilon_k$, $\varepsilon_k \overset{iid}{\sim}$ Gumbel. Expected maximum (log-sum-exp):

$$\mathbb{E}\max_k\{\delta_k + \sigma\varepsilon_k\} = \sigma\gamma_E + \sigma\ln\sum_k e^{\delta_k/\sigma}$$

For $\delta_k \ll \sigma$: $\ln\sum_k e^{\delta_k/\sigma} \approx \ln K + \frac{1}{K}\sum_k \frac{\delta_k}{\sigma}$, so

$$\mathbb{E}\max \approx \sigma(\gamma_E + \ln K) + \frac{1}{K}\sum_k \frac{s^k_{ni}}{\theta_k}$$

- the same constant appears for $i$'s exposure to $n \Rightarrow$ it cancels in the Nash difference: $\Delta^{\text{targeted}}_{ni} \approx \frac{1}{K}\Delta_{ni}$ — "isomorphic up to scale"
- this expansion is our reconstruction of the paper's one-sentence claim
- for large $\delta/\sigma$ the max dominates $\Rightarrow$ the empirical "max-sector" measures (E.2, E.4; corr. .68–.81 with FE)
- $\gamma_E \approx 0.5772$ (Euler–Mascheroni)

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# T8 Where do the trade elasticities $\theta_k$ come from?

**Broda & Weinstein (2006)**, following **Feenstra (1994)**: CES demand + upward-sloping export supply (inverse elasticity $\omega_g$) for each HTS good $g$ in US imports. Difference over time and relative to a reference supplier $r$ to remove the price index:

$$\underbrace{(\Delta^r \ln p_{gct})^2}_{Y} = \vartheta_1 \underbrace{(\Delta^r \ln s_{gct})^2}_{X_1} + \vartheta_2 \underbrace{\Delta^r \ln p_{gct}\, \Delta^r \ln s_{gct}}_{X_2} + u_{gct}$$

$$\vartheta_1 = \frac{\omega_g}{(1+\omega_g)(\sigma_g - 1)}, \quad \vartheta_2 = \frac{1-\omega_g(\sigma_g - 2)}{(1+\omega_g)(\sigma_g - 1)}$$

Identification from heteroskedasticity across supplier countries; GMM with country effects; grid search when infeasible.

- model's trade elasticity: $\theta_g = \sigma_g - 1$
- paper: HTS → simple average to HS-6 → trade-weighted (2001–21) to HS sections
- **Caliendo–Parro (2015)**: tariff variation, "tetrads" (ratios over three countries cancel unobservables)
- **Fontagné–Guimbard–Orefice (2019)**: HS-6 elasticities from bilateral applied tariffs
- **caveat**: US-based, noisy, possibly endogenous ⇒ paper reports unweighted versions everywhere

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# T9 Two-way fixed effects, dyadic data and clustering

**Frisch–Waugh–Lovell**: $\hat{\beta}$ = OLS of $\ddot{y}$ on $\ddot{x}$ after sweeping out pair and year effects (and controls).

**Unbalanced panels**: no closed-form two-way demeaning $\Rightarrow$ *alternating projections*: demean by pair, then by year, repeat to convergence (Guimarães–Portugal 2010; `reghdfe`, Correia 2016).

**Dyads**: both directions $(n, i)$ and $(i, n)$ appear. Duplication leaves $\hat{\beta}$ unchanged; clustering by *unordered* pair makes s.e. identical to one row per pair (fn. 24), while allowing coercer/target FE.

**Two-way clustering** (Cameron–Gelbach–Miller 2011):

$$\hat{V} = \hat{V}_{\text{pair}} + \hat{V}_{\text{year}} - \hat{V}_{\text{pair}\cap\text{year}}$$

(Panel C.3: s.e. $0.016 \to 0.036$, still $p < 0.01$.)

**Not done, worth trying**: *dyadic-robust* s.e. (Aronow–Samii–Assenova 2015): pairs sharing a country (US–CHN, US–JPN) may have correlated shocks.

```
reghdfe eng L.pow L.gdp L.trade, absorb(pair year) vce(cluster upair)
```

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# T10 What does a coefficient mean after within-pair standardisation?

$\tilde{x}_{pt} = (x_{pt} - \bar{x}_p)/\mathrm{sd}_p(x)$. With $x, y$ both standardised and no other regressors:

$$\hat{\beta} = \frac{\sum_p \sum_t \tilde{x}_{pt}\tilde{y}_{pt}}{\sum_p \sum_t \tilde{x}_{pt}^2} = \frac{\sum_p (T_p - 1)\,\rho_p}{\sum_p (T_p - 1)}$$

$\Rightarrow$ a (length-weighted) **average within-pair correlation**.

- every pair counts equally, however volatile its raw power
- pair FE become nearly redundant (means are zero) — why country FE (C.2) give the same $0.236$

**Reading the paper's magnitudes**

- engagement $=$ *sum* of $\approx 15$ per-bin $z$-scores, so its within-pair SD exceeds 1: $0.236$ is in *index* units
- standardising the total count instead gives $0.035$ (D.1): a $1$ SD rise in power $\approx 0.035$ SD more total events
- % effects need each pair's mean and SD; the paper reports $+38.4\%$ (mean) and $+18.8\%$ (median)
- raw-unit versions: OLS $105.3$, IV $531.0$ (F.1)

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# T11 ICEWS events and the engagement index

**ICEWS**: automated coding of world news into "who did what to whom" (BBN ACCENT coder), CAMEO event ontology; each type has an intensity on a Goldstein-type scale $[-10, 10]$ (Table A.1).

**Index construction**

1. state-to-state bilateral events, both directions merged
2. keep intensity $[-7, 8)$: non-violent $(92\%)$
3. count $N_{p,b,t}$ by pair $p$, unit bin $b$, year $t$
4. $z_{p,b,t} = \left(N_{p,b,t} - \overline{N}_{p,b}\right)/\mathrm{sd}_p(N_{p,b})$
5. $\mathrm{Engagement}_{p,t} = \sum_b z_{p,b,t}$

**Why standardise within pair?** Media coverage differs by pair (US–China news is abundant), so cross-pair levels are unreliable; changes within a pair are informative.

**Variants** (Table A.9, Panel D): standardise the total (D.1, $0.035$); all intensities $[-10, 10]$ (D.2, $0.269$); de-duplicate same-story records at pair×type×day (D.3, $0.238$).

**Caveat**: intro speaks of "19 million events"; Sec. 4.1 counts 6,733,036 bilateral events (2001–21) — different stages of filtering.

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# T12 Binned scatter plots with controls (`binsreg`)

**Goal**: show $\mathbb{E}[y \mid x, \text{controls}]$ without assuming linearity.

**Common but flawed**: residualise $y$ and $x$ on controls, then bin. Cattaneo, Crump, Farrell & Feng (2024) show this generally mis-represents the conditional mean (residualising distorts $x$).

`binsreg`: partition $x$ into $J$ quantile bins and estimate the semi-linear model

$$y_{pt} = \sum_{j=1}^{J} \mathbf{1}\{x_{pt} \in \mathcal{B}_j\} b_j + w_{pt}'\gamma + \alpha_p + \gamma_t + \varepsilon_{pt}$$

plot $\hat{b}_j$ at bin centres; $J$ by an IMSE-optimal rule; confidence bands available.

- paper: Fig. 2 (standardised) looks linear; Fig. A.7 (raw power) shows mild concavity
- concavity is what T2 predicts at high dependence — but HS sections rarely reach it
- `binsreg eng pow gdp trade, absorb(pair year)` (Stata/R/Python)

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# T13 Shift-share IV: two ways to justify it

Generic instrument $B_\ell = \sum_k z_{\ell k}\, g_k$ (shares $\times$ shifts).

**(a) Exogenous shares** (Goldsmith-Pinkham, Sorkin & Swift 2020): Bartik $=$ GMM with the $K$ share instruments; need shares (conditionally) exogenous. Diagnostics: Rotemberg weights, balance, over-ID (T14–T17).

**(b) Exogenous shocks** (Borusyak, Hull & Jaravel 2022): shares may be endogenous; identification from many as-good-as-random shocks with dispersed exposure; equivalent shock-level regression; AKM inference (T18).

**This paper**: shares $w^{\theta}_{nk\tau}$ (21 sectors, 1995–99); shifts $\hat{s}_{ikt}$ (exporter $\times$ sector $\times$ year global market shares); argued and diagnosed in the GPSS (shares) tradition.

- threat: 1995–99 import mix correlated with *later* diplomatic trends (e.g. machinery importers on different paths); pair FE absorb levels, year FE global trends
- $|b_{nit} - b_{int}|$ is non-linear in shocks $\Rightarrow$ Rotemberg and AKM use the linear, directed version (F.2)

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# T14 Rotemberg weights: which instruments drive $\hat{\beta}$?

Let $Z_k = z_k \circ g_k$ (sector-$k$ component), $B = \sum_k Z_k$; $\perp$ = after partialling out controls and FE. Then

$$\hat{\beta}_{\text{Bartik}} = \frac{B'Y^{\perp}}{B'X^{\perp}} = \sum_k \underbrace{\frac{Z_k'X^{\perp}}{\sum_{k'} Z_{k'}'X^{\perp}}}_{\hat{\alpha}_k} \underbrace{\frac{Z_k'Y^{\perp}}{Z_k'X^{\perp}}}_{\hat{\beta}_k}$$

- $\sum_k \hat{\alpha}_k = 1$; weights can be negative
- $\hat{\alpha}_k$ = sensitivity of $\hat{\beta}$ to mis-specification of instrument $k$ $\Rightarrow$ where exogeneity matters most

**Here**: machinery .38, textiles .14, plastics .09, misc. manufactures .08, chemicals .07; minerals $-.018$.

**Code sketch** (Python):

```
Xp, Yp, Zp = [resid(v, C) for v in (x, y, Z)]
a = Zp.T @ Xp # K-vector
alpha = a / a.sum()
beta_k = (Zp.T @ Yp) / a
```

Stata: `bartik_weight` (GPSS).

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# T15 Balance tests and leave-one-out shifters

**Balance** (Table A.7): for each high-weight sector $k$ and country characteristic $X$:

$$w_{nk\tau} = b_0 + b_1 \, \Delta X_n + e_{nk}$$

$\Delta X_n$ = change in within-country standardised $X$ from 1995–99 to 2017–21. $X$: WDI economy and openness, SIPRI military, CNTS politics (demonstrations, strikes, crises, ...), and $z$-score indices.

Finding: moderate, mixed-sign associations in the top-5 sectors; muted elsewhere ⇒ drop top-5: IV $= 0.647$ $(0.068)$.

**Leave-one-out shifter** (Autor, Dorn & Hanson 2013):

$$b'_{nit} = \sum_k \hat{s}_{(-n)ikt} \, w^{\theta}_{nk\tau}$$

$\hat{s}_{(-n)ikt}$ = $i$'s world market share in $k$ computed *without* sales to $n$ (e.g. China's share without the US) — removes the mechanical link between the shifter and $n$'s own demand.

Result: first stage $0.220$ (F $= 3{,}913$); IV $0.703$ $(0.060)$.

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# T16 Estimators with many instruments

**k-class** (controls partialled out, $M_Z = I - P_Z$):

$$\hat{\beta}(\kappa) = \left[X'(I - \kappa M_Z)X\right]^{-1} X'(I - \kappa M_Z)y$$

- $\kappa = 0$: OLS; $\kappa = 1$: TSLS (biased toward OLS when $K$ is large)
- **LIML**: $\kappa =$ smallest root of $\det(\bar{Y}'\bar{Y} - \kappa\, \bar{Y}' M_Z \bar{Y}) = 0$, $\bar{Y} = [y, X]$
- **Fuller**: $\kappa = \kappa_{\text{LIML}} - c/(n - K)$, finite moments
- **MBTSLS** (Kolesár et al. 2015): $\kappa = \frac{1 - L/n}{1 - K/n - L/n}$ ($K$ excluded IVs, $L$ exogenous regressors)

**HFUL** (Hausman et al. 2012): jackknife — delete own-observation terms $P_{ii}$, robust to heteroskedasticity:

$$\hat{\beta} = \left[X'PX - \textstyle\sum_i P_{ii} X_i X_i' - \tilde{\alpha} X'X\right]^{-1} \times \left[X'Py - \textstyle\sum_i P_{ii} X_i y_i - \tilde{\alpha} X'y\right]$$

**Paper** (21 instruments): TSLS $480.1$, MBTSLS $480.6$, LIML $485.5$, HFUL $492.4$ vs. Bartik $431.8$ — close.

Stata `ivreg2` (`liml`, `fuller()`, `kclass()`); R `ManyIV`.

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# T17 Over-identification and weak-instrument-robust tests

**Sargan / Hansen** $J$: $J = nR^2$ of residuals on instruments (robust GMM version) $\sim \chi^2_{K-p}$ under "all instruments valid and identify the same $\beta$".

**Anderson–Rubin**: test $H_0 : \beta = \beta_0$ by regressing $y - X\beta_0$ on $Z$ (F-test). Robust to weak IVs; with $K > 1$ it jointly tests $\beta_0$ and over-ID $\Rightarrow$ an empty confidence set signals heterogeneity/invalidity.

**CHNSW** (Chao, Hausman, Newey, Swanson & Woutersen 2014): jackknife (leave-own-out) Sargan-type statistic, standardised; valid with many instruments and heteroskedasticity.

**First-stage strength**: F $= 3{,}996$–$7{,}322 \gg$ Stock–Yogo's 10 or the Montiel Olea–Pflueger effective-F threshold.

**Reading the paper**: AR and CHNSW *reject* homogeneity across the 21 sector instruments. Under heterogeneous effects this is expected (each $\hat{\beta}_k$ is a different weighted average); the GPSS diagnosis is that high-weight, high-F sectors sit near the Bartik estimate (Fig. A.9).

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# T18 Shift-share-robust standard errors (Adão–Kolesár–Morales 2019)

**Problem**: units with similar shares load on the same unobserved sector shocks ⇒ residuals correlated across units ⇒ conventional s.e. over-reject (severely, in AKM's placebo exercises).

**Derivation** (just-identified, $B_\ell = \sum_s w_{\ell s}\xi_s$):

$$\hat\beta - \beta = \frac{\sum_\ell \ddot B_\ell \varepsilon_\ell}{\sum_\ell \ddot B_\ell X_\ell}, \quad \sum_\ell \ddot B_\ell \varepsilon_\ell \approx \sum_s \tilde\xi_s \underbrace{\sum_\ell w_{\ell s}\varepsilon_\ell}_{R_s}$$

$$\hat V_{\text{AKM}} = \frac{\sum_s \tilde\xi_s^2 \hat R_s^2}{\left(\sum_\ell \ddot B_\ell X_\ell\right)^2}, \quad \hat R_s = \sum_\ell w_{\ell s}\hat\varepsilon_\ell$$

- $\tilde\xi_s$: coefficients from regressing $\ddot B$ on the share matrix $W$; AKM0 imposes $H_0$ (better size)
- **paper**: clusters on 21 sectors: $431.8$ (s.e. $12.5$) vs. pair-clustered $64.7$
- AKM normally *widens* intervals; a smaller s.e. with only 21 shock clusters deserves a second look (AKM0, wild bootstrap)
- R: `ShiftShareSE` (`reg_ss`, `ivreg_ss`)

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# T19 Sector-specific power, product complexity and critical goods

**Two-step design**

1. for each sector $k$ (1,241 HS-4; 21 sections; 18 CP sectors) treat $k$ as the whole universe, compute *unweighted* power, re-run eq. (12) $\Rightarrow \hat{\beta}_k$
2. regress $\hat{\beta}_k$ on traits: elasticity (BW/CP/FGO), PCI, critical indicator or share; robust s.e.

**Caveats**: $\hat{\beta}_k$ is a generated outcome with heterogeneous precision $\Rightarrow$ weight by $1/\mathrm{se}_k^2$ (meta-regression); $N = 18$–$21$ at aggregate levels.

**PCI** (Hidalgo–Hausmann 2009): $M_{cp} = \mathbf{1}\{\mathrm{RCA}_{cp} \geq 1\}$, diversity $k_c = \sum_p M_{cp}$, ubiquity $k_p = \sum_c M_{cp}$,

$$\tilde{M}_{pp'} = \sum_c \frac{M_{cp} M_{cp'}}{k_c \, k_p}$$

PCI = standardised eigenvector of $\tilde{M}$ for its 2nd-largest eigenvalue (limit of the "method of reflections").

**Critical goods**: ITA Draft List of Critical Supply Chains (EO 14017, 2021): 2,409 HTS-8/10 lines, 4 sectors, 22 subsectors $\rightarrow$ HS-6 critical if any line is $\rightarrow$ HS-4 indicator (35% of 2015 world trade).

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# T20 Placebo by random draws: is the EU special?

1. EU-15 members (pre-2000) form $15 \cdot 14/2 = 105$ pairs; eq. (12) on them: $-0.046$ (s.e. $0.179$); IV $-0.222$ $(0.458)$
2. draw 105 random (EU, non-EU) pairs; re-estimate; repeat 500 times
3. compare: where does the EU estimate fall in the simulated distribution? (Fig. A.10: left tail; draws centre $\approx 0.2$)

- logic of **randomisation inference**: empirical $p \approx$ share of draws $\leq$ EU estimate
- **caveat**: random EU–non-EU pairs are not exchangeable with EU pairs (geography, trade intensity) — a descriptive benchmark, not a formal test
- interpretation: institutions with rules on bilateral engagement can switch off trade power — and the measure picks up *international*, not intranational, forces

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# T21 UNGA ideal points: dynamic ordinal IRT (BSV 2017)

Country-year position $\theta_{it}$ on one dimension (support for the US-led liberal order). For resolution $v$:

$$Z_{itv} = \beta_v \theta_{it} + \varepsilon_{itv}, \quad \varepsilon \sim N(0,1)$$

$$\text{vote} = \begin{cases} \text{no} & Z < \gamma_{1v} \\ \text{abstain} & \gamma_{1v} \le Z < \gamma_{2v} \\ \text{yes} & Z \ge \gamma_{2v} \end{cases} \quad \text{(ordered probit)}$$

$\beta_v$ = discrimination: a resolution condemning hunger ($\beta_v \approx 0$) says nothing about blocs.

- **bridging**: resolutions identical across years share $(\beta_v, \gamma_v) \Rightarrow$ positions comparable over time
- **estimation**: Bayesian MCMC with data augmentation (Albert–Chib; Johnson–Albert 1999, ch. 5); posterior means; absences treated as missing at random
- paper: pairwise similarity from these positions; regress on lagged power with lagged similarity
- not used for alignment in Sec. 5: UNGA votes partly concern trade itself

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# T22 Statistical multilateration of political positions (App. F)

**Anchors** (each year): CHN $(0,0)$; USA $(\zeta,0)$, $\zeta =$ US disapproval of China; RUS: $\alpha =$ RUS–CHN distance, $\beta =$ mean of US/RUS mutual disapproval:

$$r_1 = \frac{\alpha^2+\zeta^2-\beta^2}{2\zeta}, \quad r_2 = \sqrt{\alpha^2 - r_1^2}$$

(if the triangle inequality fails — 2009, 2010 — midpoint: $r_1 = \frac{\zeta-\beta+\alpha}{2}$, $r_2 = 0$).

**Others**: $\hat{d}_{ict} = \|(x,y) - c\| + \epsilon$, $\epsilon \sim N(0, \sigma_t^2)$:

$$g_{it}(x,y) \propto \prod_c \phi\Big(\frac{\hat{d}_{ict} - d_{ict}(x,y)}{\sigma_t}\Big), \quad (x,y) \in [-1.5, 1.5]^2$$

(uniform prior, so posterior $\propto$ likelihood).

**Noise variance**: ML at each country's best location:

$$\hat{\sigma}_t^2 = \frac{1}{3N_t} \sum_i \sum_c \big(\hat{d}_{ict} - d^*_{ict}\big)^2$$

**Caveat (Neyman–Scott)**: 3 distances but 2 free coordinates per country $\Rightarrow$ about 1 residual degree of freedom $\Rightarrow$ $\hat{\sigma}_t^2$ biased down by $\approx 3\times$; a d.o.f.-corrected version divides by $N_t$. Posteriors may be too tight. *(Our observation.)*

**Data**: Gallup 2006–2024 (consistent from 2011); missing US data (2008, 2012) imputed by Lasso from Pew; Polity carried forward from 2018.

back to talk

# T23 Rejection sampling and $k$-means compression

**Rejection sampling** from $g_{it}$ on $\Omega = [-1.5, 1.5]^2$:

1. draw $u \sim \mathrm{Unif}(\Omega)$
2. accept with probability $L_{it}(u)/L_{\max}$
3. repeat until 100,000 accepted draws

Accepted draws are exact draws from $g_{it}$; acceptance rate $= \int L / (|\Omega| L_{\max})$.

**Compression**: $k$-means with 1,000 clusters; atoms = centroids, weights = cluster shares.

**Why $k$-means?** Mapping each draw to its centroid is a feasible transport plan, so

$$W_2^2(\hat{\mu}_{\text{master}}, \hat{\mu}_{\text{compressed}}) \le \frac{1}{n} \sum_{m} \| x_m - c(x_m) \|^2$$

$=$ the $k$-means objective $\Rightarrow$ $k$-means minimises an *upper bound* on the approximation error (beats quantile binning, as the paper found).

```
km = KMeans(n_clusters=1000).fit(draws)
atoms = km.cluster_centers_
w = np.bincount(km.labels_)/len(draws)
```

back to talk

# T24 Wasserstein-2 distance and optimal transport

For $\mu = \sum_a p_a \delta_{x_a}$, $\nu = \sum_b q_b \delta_{y_b}$:

$$W_2^2(\mu,\nu) = \min_{T \geq 0} \sum_{a,b} T_{ab} \|x_a - y_b\|^2$$

$$\text{s.t.} \quad \sum_b T_{ab} = p_a \;\; \forall a, \qquad \sum_a T_{ab} = q_b \;\; \forall b$$

A linear program ($10^6$ variables for $1{,}000 \times 1{,}000$ atoms) solved exactly by network simplex.

```
M = ot.dist(X, Y) # squared Euclidean
W2 = np.sqrt(ot.emd2(p, q, M))
```

Closed form for Gaussians:
$W_2^2 = \|m_1 - m_2\|^2 + \operatorname{tr}\big(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}\big)$.

**Why not KL divergence?**

- $\mathrm{KL}(\mu\|\nu) = \sum \mu \log(\mu/\nu) = \infty$ when supports differ — always, for point clouds
- asymmetric; blind to geometry: two disjoint clouds $0.1$ or $10$ apart have the same (infinite) KL
- $W_2$ is a true metric and grows with how far mass must move

Anchors are point masses: $W_2 =$ root-mean-square distance of $i$'s cloud from the anchor.

Final: $\text{Alignment} = 1 - [\alpha d^G + (1-\alpha) d^P]$, normalised distances, $\alpha = 0.5$.

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# T25 Close-election IV with an interacted ("trend-break") first stage

**2SLS**: endogenous $a_{\{t+1,t+2\}}$ (alignment of AB); excluded instruments (as written in eqs. 18–19):

$$\mathrm{TO}_{A,t}, \quad \mathrm{TO}_{A,t} \times a_{t-1}, \quad \mathrm{NT}_{A,t}, \quad \mathrm{NT}_{A,t} \times a_{t-1}$$

included: $\mathrm{Alignment}_{t-1}$, GDP gap, trade, $\mathrm{Power}_{B\to A,t-1}$ (all $t-1$), pair & year FE.

**Why interact?** Turnover moves alignment in opposite directions depending on $\mathrm{Alignment}_{t-1}$, so the main effect averages $\approx 0$ (Table A.20); the interaction captures the *reversal*.

```
ivreghdfe powBA_post (ali_post = TO TOxali NT
NTxali) ali_pre gdp_pre trd_pre powBA_pre,
absorb(pair year) cluster(pair)
```

**Assumption**: within close races, turnover is as good as random (RD logic: Lee 2008; Marx–Pons–Rollet 2024) and affects $\mathrm{Power}_{B\to A}$ only through alignment.

**Not a sharp RD**: no running-variable polynomial; "close" $=$ |margin| $< 5\%$ plus parliamentary turnovers with negative margins; robust to 4/6/8% and presidential only.

**Inference**: shocks are election-level (13 turnovers) but s.e. are pair-clustered $\Rightarrow$ clustering by A or a wild cluster bootstrap would be a useful check.

back to talk

# T26 Multiple testing: false discovery rate (BH, BKY)

$\approx 40$ balance regressions $\Rightarrow$ some $p < 0.05$ by chance. Bonferroni (FWER) is very conservative; **FDR** $= \mathbb{E}[\text{false rejections/rejections}]$.

**Benjamini–Hochberg (1995)**: sort $p_{(1)} \le \cdots \le p_{(m)}$; reject $H_{(1)}, \ldots, H_{(k^*)}$ with

$$k^* = \max\{k : \; p_{(k)} \le kq/m\}$$

**BKY (2006), two-stage adaptive**:

1. BH at $q' = q/(1+q) \to r_1$ rejections
2. if $0 < r_1 < m$: $\hat{m}_0 = m - r_1$; BH at $q^* = q' m/\hat{m}_0$

**Sharpened $q$-value** (Anderson 2008): the smallest $q$ at which a hypothesis is rejected (grid search over $q$).

**Paper**: Table A.19 stars use BKY-adjusted $q$-values; **no** pair-level characteristic predicts turnover.

**Caveat**: with only 33 close elections, power to detect imbalance is limited — "not significant after FDR" is weak evidence of balance; several nominal $t$-statistics are $\approx 2$–$2.6$ (trade/GDP, riots, taxes on trade, arms exports).

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# T27 Eora imputations: domestic spending and services (App. C)

**Domestic expenditure** (Eora26: 184 countries, 26 sectors; free to 2017, 2015–17 average used):

$$V_{nng} = \Big( \sum_{i \in \text{For}} V^{\text{BACI}}_{nig} \Big) \times \frac{V^{\text{Eora}}_{nnk}}{\sum_{i \in \text{For}} V^{\text{Eora}}_{nik}}, \quad k = k(g)$$

then rescale so national totals match Eora:
$V^{\text{norm}}_{nng} = V_{nng} V_{nn} / \sum_g V_{nng}$.
Liberia, Madagascar, Myanmar, Somalia: Eora/BACI ratio extreme in $\geq 20$ of 21 years $\Rightarrow$ corrected with yearly median ratios.

**Services**: non-goods-to-goods ratio within each pair

$$\gamma_{ni} = \frac{\sum_{k \in \text{non-goods}} V^{\text{Eora}}_{nik}}{\sum_{k \in \text{goods}} V^{\text{Eora}}_{nik}}, \quad V_{ni,\text{serv}} = \gamma_{ni} \sum_{k \in \text{BACI}} V_{nik}$$

one extra "sector" with the average $\theta$. Non-goods = Eora sectors where BACI covers $< 10\%$ of Eora trade (construction, finance, transport, retail, ...).

Correlation with baseline (pair & year FE): $.643$ (domestic), $.846$ (services).

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# T28 Trade policies from the Global Trade Alert

**GTA**: 56,497 trade-related interventions, 225 countries, 2008–2023 (tariffs, subsidies, localisation rules, . . . ), each evaluated as liberalising or restrictive.

**Targeted policies**: keep acts whose *State Act Title* names a target country (text-based, avoids GTA's broad "affected" lists) ⇒ 2,485 policies; count new policies (flows).

**Outcomes** ($B \to A$, averaged over $t{+}1, t{+}2$):

- NetLibImp = #liberal − #restrictive import measures
- NetLibExp = same for export measures
- PowerPromoting = NetLibExp − NetLibImp (higher ⇒ more power)

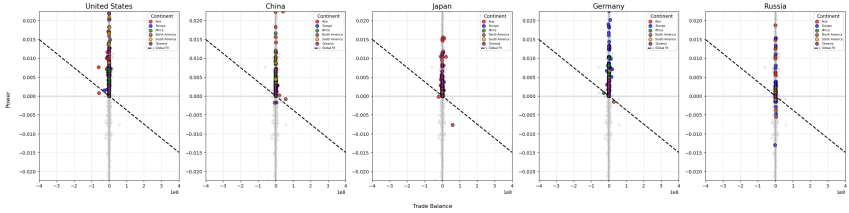
**Results** (coef. on predicted alignment): power-promoting $-0.380^{**}$; import $0.356^{***}$; export $-0.0104$ (n.s.). Placebo A→B: all n.s. (Table A.25). Sample $N = 4{,}861$.

back to talk

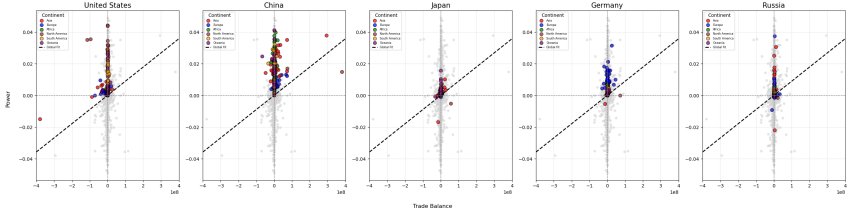
# Backup II

**Additional results from the appendix**

# Power vs. trade balance (Figure A.2)



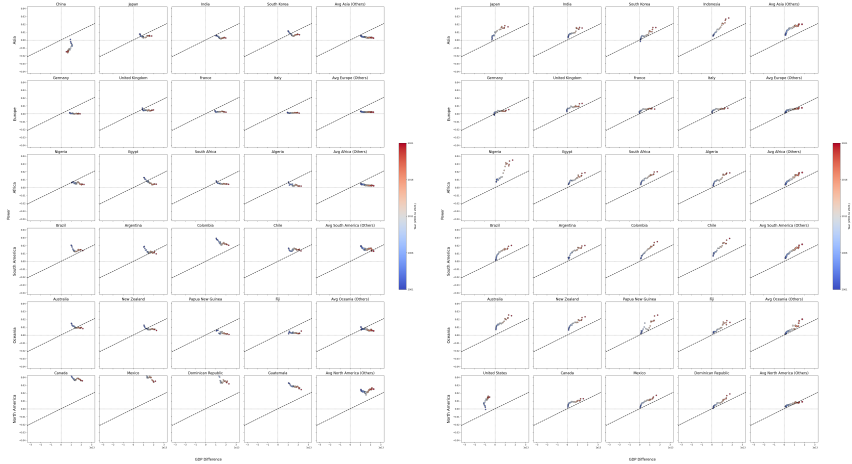

Panel A: 2001




Panel C: 2021. Coercers: US, China, Japan, Germany, Russia; dotted line = global fit.

Most coercer–target-years have a trade balance near zero, yet power varies widely: power $\neq$ imbalance.

# Power vs. GDP gap: the US and China (Figure A.3)




Left: US; right: China. Rows: continents; dots: years (blue 2001 → red 2021); dashed: global fit of power on GDP gap.

China sits *above* the fit almost everywhere (Asia, South America), even with a negative GDP gap.

The US sits *below* the fit in Asia, Europe, Africa, Oceania — but *above* in the Americas.

# Engagement data: distribution and the Italy–Russia timeline

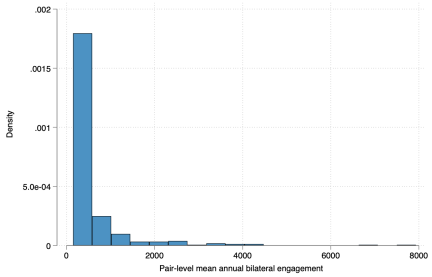



Figure A.5: pair-level mean annual events, pairs with mean $> 150$.

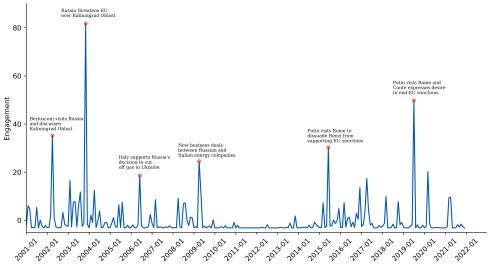



Figure A.4, Panel B: Italy–Russia (2006 gas dispute; 2015 Putin visit to discourage EU sanctions).

Engagement can be cooperative or adversarial; the paper therefore studies the *amount* of engagement, not its direction.

# Non-standardised binscatter and the source of shift-share variation

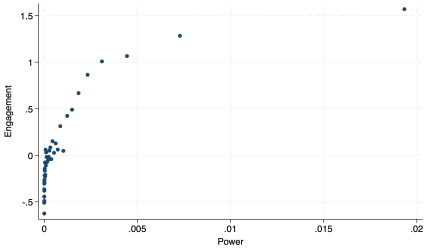



Figure A.7: raw (non-standardised) power: mild concavity.

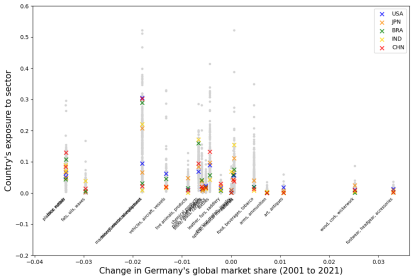
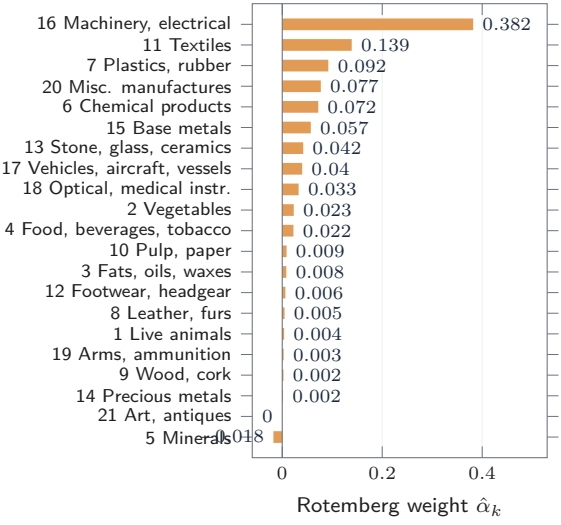



Figure A.8: change in Germany's global market share by sector (x) vs. importers' 1995–99 exposure (y).

Within a sector (vertical strip), importers differ in exposure; across sectors, Germany's market share moves differently — the product generates the instrument.

# All Rotemberg weights (Table A.6)




- weights sum to one; two are (slightly) negative (minerals, art)
- top five (machinery, textiles, plastics, misc. manufactures, chemicals) $\approx 0.76$
- computed on the directed, non-standardised specification (F.2), where the instrument is linear in share $\times$ shift
- balance tests (Table A.7) focus on the top eight

# Sector-level coefficients (Figures A.11, A.12)

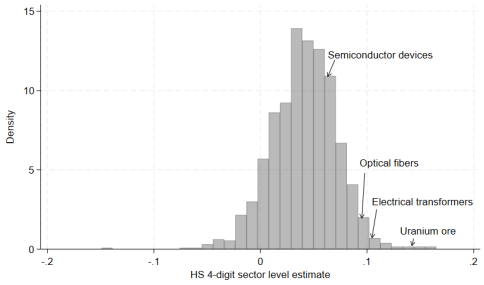



Figure A.11: 1,241 HS-4 sector coefficients.

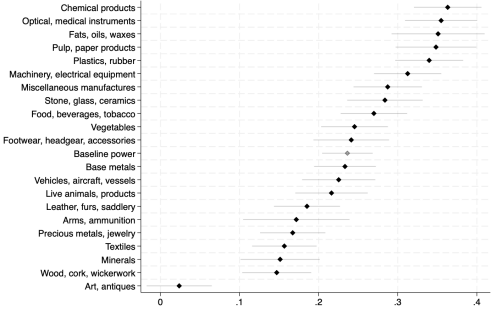



Figure A.12: HS-section coefficients (controls for within-sector trade).

The text names chemicals, optical/medical instruments and machinery as the top three sections; in Fig. A.12 fats/oils, pulp/paper and plastics also rank above machinery — the top six are statistically indistinguishable.

# Critical goods and exporter vs. importer power (Figures A.13, A.14)

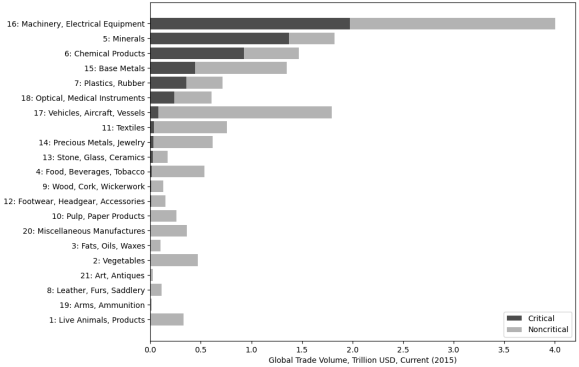



Figure A.13: 2015 world trade by HS section; dark = critical goods (ITA list).

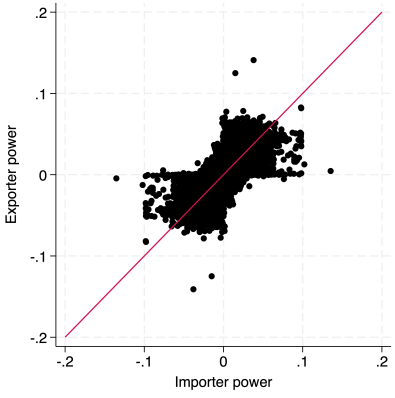



Figure A.14: pair averages of exporter vs. importer power; red = $45^{\circ}$.

# Other sources of influence (Table A.16)

| | Military expenditure | | | Foreign aid | | | Sovereign debt | | |
|---|---|---|---|---|---|---|---|---|---|
| | control | $<$ median | $>$ median | control | no aid | aid | control | no debt | debt |
| Power$_{t-1}$ | $0.244^{***}$ | $0.160^{***}$ | $0.253^{***}$ | $0.237^{***}$ | $0.221^{***}$ | $0.227^{***}$ | $0.227^{***}$ | $0.190^{***}$ | $0.285^{***}$ |
| | (0.019) | (0.024) | (0.028) | (0.016) | (0.020) | (0.028) | (0.016) | (0.017) | (0.048) |
| Other influence$_{t-1}$ | $0.048^{**}$ | | | $0.037^{*}$ | | | $0.240^{***}$ | | |
| | (0.020) | | | (0.021) | | | (0.045) | | |
| $N$ | 390,744 | 186,344 | 186,020 | 517,298 | 341,584 | 175,714 | 517,298 | 438,308 | 78,990 |

- other sources of influence predict engagement themselves, but leave the power coefficient unchanged
- power bites harder where military or financial asymmetries also exist $\Rightarrow$ **complements** (consistent with theories of hegemony, e.g. Clayton, Maggiori & Schreger 2025)

# Heterogeneity in IV form (Tables A.11, A.12)

| | OLS (col. 3) | | IV (col. 6) | |
|---|---|---|---|---|
| Sample | coef. | s.e. | coef. | s.e. |
| EU-15 members among themselves ($N = 4{,}200$) | $-0.046$ | (0.179) | $-0.222$ | (0.458) |
| Pairs with $\geq 1$ violent event | $0.303^{***}$ | (0.033) | $0.652^{***}$ | (0.107) |
| Pairs with no violent event | $0.104^{***}$ | (0.015) | $0.524^{***}$ | (0.065) |
| Pairs with $\geq 1$ trade sanction | $0.564^{***}$ | (0.113) | $1.070^{***}$ | (0.301) |
| Pairs with no sanction | $0.189^{***}$ | (0.015) | $0.618^{***}$ | (0.059) |

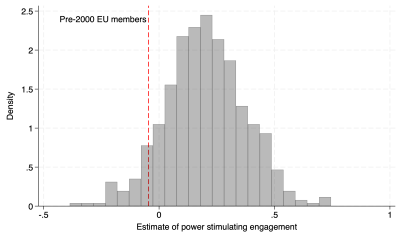



Figure A.10: 500 random samples of 105 EU–non-EU pairs; red line = EU-only estimate.

Interpretation (paper): realised violence or sanctions are off-path events that *discipline beliefs* and make trade threats more credible; EU rules neutralise bilateral leverage.

# The close elections (Table A.18)

| Country | Year | TO | Type | Margin |
|---|---|---|---|---|
| Argentina | 2015 | **1** | Pres. | 2.7 |
| Australia | 2016 | 0 | Parl. | −4.7 |
| Austria | 2013 | 0 | Parl. | −2.7 |
| Belgium | 2014 | **1** | Parl. | −1.3 |
| Bulgaria | 2013 | **1** | Parl. | −5.4 |
| Brazil | 2014 | 0 | Pres. | −3.3 |
| Denmark | 2015 | **1** | Parl. | −5.6 |
| Dominican Rep. | 2012 | 0 | Pres. | −4.3 |
| Ecuador | 2017 | 0 | Pres. | −2.3 |
| Estonia | 2015 | 0 | Parl. | −3.0 |
| France | 2012 | **1** | Pres. | 3.3 |
| Gabon | 2016 | 0 | Pres. | −1.6 |
| Ghana | 2012 | 0 | Pres. | −3.0 |
| Honduras | 2017 | 0 | Pres. | −1.5 |
| Ireland | 2016 | 0 | Parl. | −3.2 |
| Israel | 2015 | 0 | Parl. | −5.0 |
| Rep. of Korea | 2012 | 0 | Pres. | −3.5 |
| Sri Lanka | 2015 | **1** | Pres. | 3.7 |
| Lithuania | 2012 | **1** | Parl. | 3.5 |
| Luxembourg | 2013 | **1** | Parl. | −16.7 |
| Latvia | 2014 | 0 | Parl. | 1.0 |
| Moldova | 2014 | 0 | Parl. | 2.0 |
| Netherlands | 2012 | 0 | Parl. | −2.0 |
| Norway | 2017 | 0 | Parl. | 2.4 |
| New Zealand | 2017 | **1** | Parl. | −8.3 |
| Poland | 2015 | **1** | Pres. | 3.1 |
| Portugal | 2015 | **1** | Parl. | −9.1 |
| Paraguay | 2018 | 0 | Pres. | −3.9 |
| Sierra Leone | 2018 | **1** | Pres. | 3.6 |
| El Salvador | 2014 | 0 | Pres. | −0.2 |
| Sweden | 2014 | **1** | Parl. | −8.0 |
| Sweden | 2018 | 0 | Parl. | −0.3 |
| Venezuela | 2013 | 0 | Pres. | −1.5 |

33 close elections, 13 turnovers (TO $= 1$).

Margin = incumbent's vote margin (pp).

Parliamentary turnovers with large negative margins (e.g. Luxembourg, Portugal) enter because coalitions, not vote shares, decide who governs.

# First stage in the data and random A/B roles (Figures A.17, A.18)

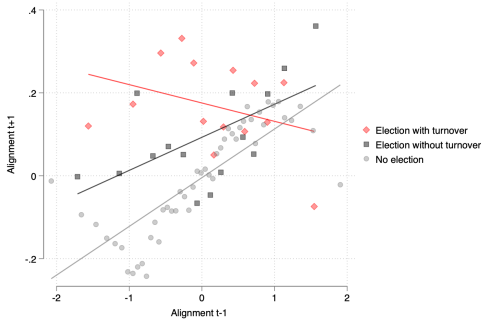



Figure A.17: alignment $t+1$ vs. $t-1$ (residualised), by election outcome.

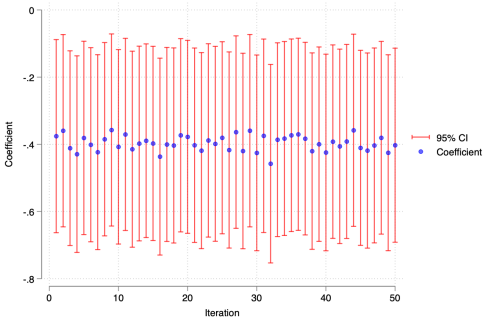



Figure A.18: second-stage coefficient for 50 random A/B designations of pairs without elections.

# Identification checks in numbers (Tables A.19, A.20)

**Table A.20: no average effect of turnover**

| | Alignment$_{\{t+1,t+2\}}$ | Power$_{\{t+1,t+2\}}$ |
|---|---|---|
| Turnover$_{A,t}$ | 0.0187 | 0.0245 |
| | (0.0268) | (0.0231) |
| $N$ | 3,485 | 3,485 |

A's turnover moves its alignment with some partners up and others down, so *on average* nothing moves — the identifying variation is the interaction with pre-election alignment.

**Table A.19: is turnover predictable?**

Turnover regressed on about 40 pairwise differences (A − B) in domestic economy, international economy, military and political conditions at $t-1$, one at a time; s.e. clustered by coercer; stars use BKY sharpened $q$-values.

Result: **none** significant after FDR adjustment; summary $z$-scores: domestic $0.007$, international $-0.050$, military $0.074$, political $-0.034$.